

Supplementary materials for the paper “Matrix-valued Factor Model with Time-varying Main Effects” - Proof of all theorems and lemmas

1 Proof of Theorems

Proof of Theorem 1. Suppose we have another set of parameters, $(\tilde{\mu}_t, \tilde{\alpha}_t, \tilde{\beta}_t, \tilde{\mathbf{Q}}_r, \tilde{\mathbf{Q}}_c, \tilde{\mathbf{F}}_{Z,t})$ for $t \in [T]$, also satisfying (3.1). For each $t \in [T]$, left-multiplying by $\mathbf{1}'_p$ and right-multiplying by $\mathbf{1}_q$ on (3.1), we would arrive at $pq\tilde{\mu}_t = pq\mu_t$ according to (IC1), so μ_t is identified. Similarly, $\tilde{\alpha}_t = \alpha_t$ and $\tilde{\beta}_t = \beta_t$, by separately left-multiplying $\mathbf{1}'_p$ and right-multiplying $\mathbf{1}_q$ on $\mathbf{Y}_t$.

We hence have $\tilde{\mathbf{Q}}_r \tilde{\mathbf{F}}_{Z,t} \tilde{\mathbf{Q}}'_c = \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}'_c$, but the factor loading matrices and factor series require further identification due to the multiplicative form. Without loss of generality, write

$$\tilde{\mathbf{Q}}_r = \mathbf{Q}_r \mathbf{M}_r + \mathbf{\Gamma}_r, \quad \text{where } \mathbf{\Gamma}'_r \mathbf{Q}_r = \mathbf{0},$$

with $\mathbf{M}_r \in \mathbb{R}^{k_r \times k_r}$ and $\mathbf{\Gamma}_r \in \mathbb{R}^{p \times k_r}$, but can have zero columns. Then we have

$$\mathbf{0} = \mathbf{\Gamma}'_r \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}'_c \tilde{\mathbf{Q}}_c = \mathbf{\Gamma}'_r \tilde{\mathbf{Q}}_r \tilde{\mathbf{F}}_{Z,t} \tilde{\mathbf{Q}}'_c \tilde{\mathbf{Q}}_c = \mathbf{\Gamma}'_r \mathbf{\Gamma}_r \tilde{\mathbf{F}}_{Z,t} \tilde{\mathbf{Q}}'_c \tilde{\mathbf{Q}}_c,$$

which can only be true in general if $\mathbf{\Gamma}_r = \mathbf{0}$ since $\tilde{\mathbf{F}}_t$ is random and $\tilde{\mathbf{Q}}'_c \tilde{\mathbf{Q}}_c \rightarrow \Sigma_{A,c}$ due to (L1). Using (L1), $\mathbf{M}_r$ is of full rank and hence $\tilde{\mathbf{Q}}_r$ and $\mathbf{Q}_r$ share the same column space. Similarly, the factor loading space of $\mathbf{Q}_c$ is identified, and $\mathbf{F}_{Z,t}$ is hence identified once $\mathbf{Q}_r$ and $\mathbf{Q}_c$ are given correspondingly. $\square$

Proof of Theorem 2. By Assumption (IC1), we have $\mu_t = \mathbf{1}'_p (\mathbf{Y}_t - \mathbf{E}_t) \mathbf{1}_q / pq$ and hence

$$(\hat{\mu}_t - \mu_t)^2 = \frac{1}{p^2 q^2} \left(\mathbf{1}'_p \mathbf{E}_t \mathbf{1}_q \right)^2 = \frac{1}{p^2 q^2} \left(\sum_{i=1}^p \sum_{j=1}^q E_{t,ij} \right)^2. \quad (\text{S.1})$$

Assumption (E1) implies each entry of $\mathbf{E}_t$ has zero mean and bounded fourth moment, and we have

$$\mathbb{E} \left[\left(\sum_{i=1}^p \sum_{j=1}^q E_{t,ij} \right)^2 \right] = \text{Var} \left(\sum_{i=1}^p \sum_{j=1}^q E_{t,ij} \right) = \sum_{i=1}^p \sum_{l=1}^p \sum_{j=1}^q \sum_{h=1}^q \text{cov}(E_{t,ij}, E_{t,lh}) = O(pq), \quad (\text{S.2})$$

where we used Lemma 1 in the last equality. Thus with (S.1), $(\hat{\mu}_t - \mu_t)^2 = O_P(p^{-1}q^{-1})$.

Similar to the rate for $\hat{\mu}_t$, by again (IC1) we have $\alpha_t = q^{-1} \mathbf{Y}_t \mathbf{1}_q - \mu_t \mathbf{1}_p - q^{-1} \mathbf{E}_t \mathbf{1}_q$ and $\beta_t = p^{-1} \mathbf{Y}'_t \mathbf{1}_p - \mu_t \mathbf{1}_q - p^{-1} \mathbf{E}'_t \mathbf{1}_p$. Then we have

$$\frac{1}{p} \cdot \|\hat{\alpha}_t - \alpha_t\|^2 = \frac{1}{p} \cdot \left\| (\mu_t - \hat{\mu}_t) \mathbf{1}_p + q^{-1} \mathbf{E}_t \mathbf{1}_q \right\|^2, \quad (\text{S.3})$$

$$\frac{1}{q} \cdot \|\hat{\beta}_t - \beta_t\|^2 = \frac{1}{q} \cdot \left\| (\mu_t - \hat{\mu}_t) \mathbf{1}_q + p^{-1} \mathbf{E}'_t \mathbf{1}_p \right\|^2. \quad (\text{S.4})$$

From (S.1) we have $\|(\mu_t - \hat{\mu}_t) \mathbf{1}_p\|^2 = O_P(q^{-1})$ and $\|(\mu_t - \hat{\mu}_t) \mathbf{1}_q\|^2 = O_P(p^{-1})$. Furthermore, by Lemma 1,

$$\mathbb{E} \left[\|q^{-1} \mathbf{E}_t \mathbf{1}_q\|^2 \right] = q^{-2} \cdot \sum_{i=1}^p \text{Var} \left(\sum_{j=1}^q E_{t,ij} \right) = O(pq^{-1}).$$

Similarly, $\|p^{-1} \mathbf{E}'_t \mathbf{1}_p\|^2 = O_P(qp^{-1})$. Then we have (S.3) and (S.4) as

$$\frac{1}{p} \cdot \|\hat{\alpha}_t - \alpha_t\|^2 = O_P(q^{-1}), \quad \frac{1}{q} \cdot \|\hat{\beta}_t - \beta_t\|^2 = O_P(p^{-1}).$$

In the rest of the proof, we show consistency of the factor loading estimators. From (3.4), we have

$$\begin{aligned}\widehat{\mathbf{L}}_t &= \mathbf{Y}_t + (pq)^{-1} \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}'_q - q^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q - p^{-1} \mathbf{1}_p \mathbf{1}'_p \mathbf{Y}_t, \\ \widehat{\mathbf{L}}'_t &= \mathbf{Y}'_t + (pq)^{-1} \mathbf{1}_q \mathbf{1}'_p \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p - q^{-1} \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t - p^{-1} \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p,\end{aligned}$$

and hence the following decomposition

$$\begin{aligned}\widehat{\mathbf{L}}_t \widehat{\mathbf{L}}'_t &= \mathbf{Y}_t \mathbf{Y}'_t + (pq)^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_p \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p - q^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t - p^{-1} \mathbf{Y}_t \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p \\ &\quad + (pq)^{-1} \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}'_q \mathbf{Y}'_t + (pq)^{-2} \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}'_q \mathbf{1}_q \mathbf{1}'_p \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p \\ &\quad - q^{-1} (pq)^{-1} \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}'_q \mathbf{1}'_q \mathbf{Y}'_t - p^{-1} (pq)^{-1} \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p \\ &\quad - q^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t - q^{-1} (pq)^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{1}_q \mathbf{1}'_p \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p + q^{-2} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t + q^{-1} p^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p \\ &\quad - p^{-1} \mathbf{1}_p \mathbf{1}'_p \mathbf{Y}_t \mathbf{Y}'_t - (pq)^{-1} p^{-1} \mathbf{1}_p \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_p \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p + q^{-1} p^{-1} \mathbf{1}_p \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t + p^{-2} \mathbf{1}_p \mathbf{1}'_p \mathbf{Y}_t \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p \\ &= \mathbf{Y}_t \mathbf{Y}'_t + (pq)^{-1} \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_p - p^{-1} \mathbf{Y}_t \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p - q^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t - p^{-1} \mathbf{1}_p \mathbf{1}'_p \mathbf{Y}_t \mathbf{Y}'_t \\ &\quad - (pq)^{-1} p^{-1} \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p + (pq)^{-1} \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}'_q \mathbf{Y}'_t + p^{-2} \mathbf{1}'_p \mathbf{Y}_t \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p \\ &= \mathbf{Y}_t \mathbf{Y}'_t + \mathcal{Q}_1 - \mathcal{Q}_2 - \mathcal{Q}_3 - \mathcal{Q}_4 - \mathcal{Q}_5 + \mathcal{Q}_6 + \mathcal{Q}_7, \quad \text{where} \\ \mathcal{Q}_1 &:= (pq)^{-1} \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_p, \quad \mathcal{Q}_2 := p^{-1} \mathbf{Y}_t \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p, \quad \mathcal{Q}_3 := q^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t, \quad \mathcal{Q}_4 := p^{-1} \mathbf{1}_p \mathbf{1}'_p \mathbf{Y}_t \mathbf{Y}'_t, \\ \mathcal{Q}_5 &:= (pq)^{-1} p^{-1} \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p, \quad \mathcal{Q}_6 := (pq)^{-1} \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}'_q \mathbf{Y}'_t, \quad \mathcal{Q}_7 := p^{-2} \mathbf{1}'_p \mathbf{Y}_t \mathbf{Y}'_t \mathbf{1}_p \mathbf{1}'_p.\end{aligned}\tag{S.5}$$

From (3.1), we have $\mathbf{1}'_q \mathbf{Y}'_t \mathbf{1}_p = pq\mu_t + \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p$ and $\mathbf{Y}_t \mathbf{1}_q \mathbf{1}'_p = q\mu_t \mathbf{1}_p \mathbf{1}'_p + q\boldsymbol{\alpha}_t \mathbf{1}'_p + \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p$. Thus,

$$\begin{aligned}\mathcal{Q}_1 &= (pq)^{-1} (pq^2 \mu_t^2 \mathbf{1}_p \mathbf{1}'_p + pq^2 \mu_t \boldsymbol{\alpha}_t \mathbf{1}'_p + pq\mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p + q\mu_t \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_p + q \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \boldsymbol{\alpha}_t \mathbf{1}'_p + \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p) \\ &= q\mu_t^2 \mathbf{1}_p \mathbf{1}'_p + q\mu_t \boldsymbol{\alpha}_t \mathbf{1}'_p + \mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p + p^{-1} \mu_t \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_p + p^{-1} \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \boldsymbol{\alpha}_t \mathbf{1}'_p + (pq)^{-1} \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p.\end{aligned}\tag{S.6}$$

Similarly, we have $\mathbf{Y}_t \mathbf{Y}'_t = (\mu_t \mathbf{1}_p \mathbf{1}'_q + \boldsymbol{\alpha}_t \mathbf{1}'_q + \mathbf{1}_p \boldsymbol{\beta}'_t + \mathbf{C}_t + \mathbf{E}_t)(\mu_t \mathbf{1}_q \mathbf{1}'_p + \mathbf{1}_q \boldsymbol{\alpha}'_t + \boldsymbol{\beta}_t \mathbf{1}'_p + \mathbf{C}'_t + \mathbf{E}'_t)$. Further with Assumption (IC1),

$$\begin{aligned}\mathcal{Q}_2 &= p^{-1} (\mu_t^2 \mathbf{1}_p \mathbf{1}'_q \mathbf{1}_q \mathbf{1}'_p + \mu_t \mathbf{1}_p \mathbf{1}'_q \mathbf{1}_q \boldsymbol{\alpha}'_t + \mu_t \mathbf{1}_p \mathbf{1}'_q \boldsymbol{\beta}_t \mathbf{1}'_p + \mu_t \mathbf{1}_p \mathbf{1}'_q \mathbf{C}'_t + \mu_t \mathbf{1}_p \mathbf{1}'_q \mathbf{E}'_t \\ &\quad + \mu_t \boldsymbol{\alpha}_t \mathbf{1}'_q \mathbf{1}_q \mathbf{1}'_p + \boldsymbol{\alpha}_t \mathbf{1}'_q \mathbf{1}_q \boldsymbol{\alpha}'_t + \boldsymbol{\alpha}_t \mathbf{1}'_q \boldsymbol{\beta}_t \mathbf{1}'_p + \boldsymbol{\alpha}_t \mathbf{1}'_q \mathbf{C}'_t + \boldsymbol{\alpha}_t \mathbf{1}'_q \mathbf{E}'_t \\ &\quad + \mu_t \mathbf{1}_p \boldsymbol{\beta}'_t \mathbf{1}_q \mathbf{1}'_p + \mathbf{1}_p \boldsymbol{\beta}'_t \mathbf{1}_q \boldsymbol{\alpha}'_t + \mathbf{1}_p \boldsymbol{\beta}'_t \boldsymbol{\beta}_t \mathbf{1}'_p + \mathbf{1}_p \boldsymbol{\beta}'_t \mathbf{C}'_t + \mathbf{1}_p \boldsymbol{\beta}'_t \mathbf{E}'_t \\ &\quad + \mu_t \mathbf{C}_t \mathbf{1}_q \mathbf{1}'_p + \mathbf{C}_t \mathbf{1}_q \boldsymbol{\alpha}'_t + \mathbf{C}_t \boldsymbol{\beta}_t \mathbf{1}'_p + \mathbf{C}_t \mathbf{C}'_t + \mathbf{C}_t \mathbf{E}'_t \\ &\quad + \mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p + \mathbf{E}_t \mathbf{1}_q \boldsymbol{\alpha}'_t + \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}'_p + \mathbf{E}_t \mathbf{C}'_t + \mathbf{E}_t \mathbf{E}'_t) \mathbf{1}_p \mathbf{1}'_p \\ &= q\mu_t^2 \mathbf{1}_p \mathbf{1}'_p + p^{-1} \mu_t \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_p + q\mu_t \boldsymbol{\alpha}_t \mathbf{1}'_p + p^{-1} \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \boldsymbol{\alpha}_t \mathbf{1}'_p + \mathbf{1}_p \boldsymbol{\beta}'_t \boldsymbol{\beta}_t \mathbf{1}'_p + p^{-1} \boldsymbol{\beta}'_t \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_p \\ &\quad + \mathbf{C}_t \boldsymbol{\beta}_t \mathbf{1}'_p + p^{-1} \mathbf{C}_t \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_p + \mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p + \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}'_p + p^{-1} \mathbf{E}_t \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_p.\end{aligned}\tag{S.7}$$

Since $\mathbf{Y}_t \mathbf{1}_q = q\mu_t \mathbf{1}_p + q\boldsymbol{\alpha}_t + \mathbf{E}_t \mathbf{1}_q$, we have

$$\begin{aligned}\mathcal{Q}_3 &= q^{-1} (q\mu_t \mathbf{1}_p + q\boldsymbol{\alpha}_t + \mathbf{E}_t \mathbf{1}_q)(q\mu_t \mathbf{1}'_p + q\boldsymbol{\alpha}'_t + \mathbf{1}'_q \mathbf{E}'_t) \\ &= q\mu_t^2 \mathbf{1}_p \mathbf{1}'_p + q\mu_t \mathbf{1}_p \boldsymbol{\alpha}'_t + \mu_t \mathbf{1}_p \mathbf{1}'_q \mathbf{E}'_t + q\mu_t \boldsymbol{\alpha}_t \mathbf{1}'_p + q\boldsymbol{\alpha}_t \boldsymbol{\alpha}'_t + \boldsymbol{\alpha}_t \mathbf{1}'_q \mathbf{E}'_t + \mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p + \mathbf{E}_t \mathbf{1}_q \boldsymbol{\alpha}'_t + q^{-1} \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_q \mathbf{E}'_t.\end{aligned}\tag{S.8}$$

Similar to (S.7), we have

$$\begin{aligned}
\mathcal{Q}_4 &= q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' + q\mu_t \mathbf{1}_p \boldsymbol{\alpha}_t' + \mu_t \mathbf{1}_p \mathbf{1}_q' \boldsymbol{\beta}_t \mathbf{1}_p' + \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{C}_t' + \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' + \mu_t \mathbf{1}_p \mathbf{1}_p' \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{1}_p' \\
&+ p^{-1} \mathbf{1}_p \mathbf{1}_p' \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{1}_p \boldsymbol{\alpha}_t' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \boldsymbol{\alpha}_t \mathbf{1}_q' \boldsymbol{\beta}_t \mathbf{1}_p' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{C}_t' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{E}_t' \\
&+ \mu_t \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{1}_q \mathbf{1}_p' + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{1}_q \boldsymbol{\alpha}_t' + \mathbf{1}_p \boldsymbol{\beta}_t' \boldsymbol{\beta}_t \mathbf{1}_p' + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{C}_t' + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{E}_t' + p^{-1} \mu_t \mathbf{1}_p \mathbf{1}_p' \mathbf{C}_t \mathbf{1}_q \mathbf{1}_p' \\
&+ p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{C}_t \mathbf{1}_q \boldsymbol{\alpha}_t' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{C}_t \boldsymbol{\beta}_t \mathbf{1}_p' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{C}_t \mathbf{C}_t' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{C}_t \mathbf{E}_t' \\
&+ p^{-1} \mu_t \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \boldsymbol{\alpha}_t' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}_p' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{C}_t' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \\
&= q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' + q\mu_t \mathbf{1}_p \boldsymbol{\alpha}_t' + \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' + \boldsymbol{\beta}_t' \boldsymbol{\beta}_t \mathbf{1}_p \mathbf{1}_p' + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{C}_t' + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{E}_t' \\
&+ p^{-1} \mu_t \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \boldsymbol{\alpha}_t' + p^{-1} \mathbf{1}_p' \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{C}_t' + p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t',
\end{aligned} \tag{S.9}$$

where the last equality used Assumption (IC1). For $\mathcal{Q}_5$ and $\mathcal{Q}_6$, we have

$$\begin{aligned}
\mathcal{Q}_5 &= (pq)^{-1} p^{-1} (pq\mu_t + \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q) (pq\mu_t + \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p) \mathbf{1}_p \mathbf{1}_p' \\
&= q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mu_t \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mu_t \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_p' + (pq)^{-1} p^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p',
\end{aligned} \tag{S.10}$$

and

$$\begin{aligned}
\mathcal{Q}_6 &= (\mu_t + (pq)^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p) (q\mu_t \mathbf{1}_p \mathbf{1}_p' + q \mathbf{1}_p \boldsymbol{\alpha}_t' + \mathbf{1}_p \mathbf{1}_q' \boldsymbol{\beta}_t \mathbf{1}_p' + \mathbf{1}_p \mathbf{1}_q' \mathbf{C}_t' + \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t') \\
&= q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' + q\mu_t \mathbf{1}_p \boldsymbol{\alpha}_t' + \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' + p^{-1} \mu_t \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \boldsymbol{\alpha}_t' + (pq)^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t'.
\end{aligned} \tag{S.11}$$

Lastly for $\mathcal{Q}_7$, we have similar to (S.7) that

$$\begin{aligned}
\mathcal{Q}_7 &= p^{-2} \mathbf{1}_p' (\mu_t^2 \mathbf{1}_p \mathbf{1}_q' \mathbf{1}_q \mathbf{1}_p' + \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{1}_q \boldsymbol{\alpha}_t' + \mu_t \mathbf{1}_p \mathbf{1}_q' \boldsymbol{\beta}_t \mathbf{1}_p' + \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{C}_t' + \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' \\
&+ \mu_t \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{1}_q \mathbf{1}_p' + \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{1}_q \boldsymbol{\alpha}_t' + \boldsymbol{\alpha}_t \mathbf{1}_q' \boldsymbol{\beta}_t \mathbf{1}_p' + \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{C}_t' + \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{E}_t' \\
&+ \mu_t \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{1}_q \mathbf{1}_p' + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{1}_q \boldsymbol{\alpha}_t' + \mathbf{1}_p \boldsymbol{\beta}_t' \boldsymbol{\beta}_t \mathbf{1}_p' + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{C}_t' + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{E}_t' \\
&+ \mu_t \mathbf{C}_t \mathbf{1}_q \mathbf{1}_p' + \mathbf{C}_t \mathbf{1}_q \boldsymbol{\alpha}_t' + \mathbf{C}_t \boldsymbol{\beta}_t \mathbf{1}_p' + \mathbf{C}_t \mathbf{C}_t' + \mathbf{C}_t \mathbf{E}_t' \\
&+ \mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' + \mathbf{E}_t \mathbf{1}_q \boldsymbol{\alpha}_t' + \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}_p' + \mathbf{E}_t \mathbf{C}_t' + \mathbf{E}_t \mathbf{E}_t') \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p' \\
&= p^{-2} (pq\mu_t^2 \mathbf{1}_p' + pq\mu_t \boldsymbol{\alpha}_t' + p\mu_t \mathbf{1}_q' \mathbf{E}_t' + p\boldsymbol{\beta}_t' \boldsymbol{\beta}_t \mathbf{1}_p' + p\boldsymbol{\beta}_t' \mathbf{C}_t' + p\boldsymbol{\beta}_t' \mathbf{E}_t' \\
&+ \mu_t \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' + \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \boldsymbol{\alpha}_t' + \mathbf{1}_p' \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}_p' + \mathbf{1}_p' \mathbf{E}_t \mathbf{C}_t' + \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t') \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p' \\
&= q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mu_t \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p' + \boldsymbol{\beta}_t' \boldsymbol{\beta}_t \mathbf{1}_p \mathbf{1}_p' + p^{-1} \boldsymbol{\beta}_t' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p' \\
&+ p^{-1} \mu_t \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mathbf{1}_p' \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}_p \mathbf{1}_p' + p^{-2} \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p'.
\end{aligned} \tag{S.12}$$

With (S.6), (S.7), (S.8), (S.9), (S.10), (S.11), and (S.12), we have

$$\begin{aligned}
\widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t' &= \mathbf{Y}_t \mathbf{Y}_t' + \mathcal{Q}_1 - \mathcal{Q}_2 - \mathcal{Q}_3 - \mathcal{Q}_4 - \mathcal{Q}_5 + \mathcal{Q}_6 + \mathcal{Q}_7 \\
&= q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' + q\mu_t \mathbf{1}_p \boldsymbol{\alpha}_t' + \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' + q\mu_t \boldsymbol{\alpha}_t \mathbf{1}_p' + q\boldsymbol{\alpha}_t \boldsymbol{\alpha}_t' + \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{E}_t' + \mathbf{1}_p \boldsymbol{\beta}_t' \boldsymbol{\beta}_t \mathbf{1}_p' + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{C}_t' \\
&\quad + \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{E}_t' + \mathbf{C}_t \boldsymbol{\beta}_t \mathbf{1}_p' + \mathbf{C}_t \mathbf{C}_t' + \mathbf{C}_t \mathbf{E}_t' + \mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' + \mathbf{E}_t \mathbf{1}_q \boldsymbol{\alpha}_t' + \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}_p' + \mathbf{E}_t \mathbf{C}_t' + \mathbf{E}_t \mathbf{E}_t' \\
&\quad + q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' + q\mu_t \boldsymbol{\alpha}_t \mathbf{1}_p' + \mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' + p^{-1} \mu_t \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \boldsymbol{\alpha}_t \mathbf{1}_p' + (pq)^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' \\
&\quad - q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' - p^{-1} \mu_t \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' - q\mu_t \boldsymbol{\alpha}_t \mathbf{1}_p' - p^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \boldsymbol{\alpha}_t \mathbf{1}_p' - \mathbf{1}_p \boldsymbol{\beta}_t' \boldsymbol{\beta}_t \mathbf{1}_p' - p^{-1} \boldsymbol{\beta}_t' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' \\
&\quad - \mathbf{C}_t \boldsymbol{\beta}_t \mathbf{1}_p' - p^{-1} \mathbf{C}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' - \mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' - \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}_p' - p^{-1} \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' - q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' - q\mu_t \mathbf{1}_p \boldsymbol{\alpha}_t' \\
&\quad - \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' - q\mu_t \boldsymbol{\alpha}_t \mathbf{1}_p' - q\boldsymbol{\alpha}_t \boldsymbol{\alpha}_t' - \boldsymbol{\alpha}_t \mathbf{1}_q' \mathbf{E}_t' - \mu_t \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' - \mathbf{E}_t \mathbf{1}_q \boldsymbol{\alpha}_t' - q^{-1} \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \\
&\quad - q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' - q\mu_t \mathbf{1}_p \boldsymbol{\alpha}_t' - \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' - \boldsymbol{\beta}_t' \boldsymbol{\beta}_t \mathbf{1}_p \mathbf{1}_p' - \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{C}_t' - \mathbf{1}_p \boldsymbol{\beta}_t' \mathbf{E}_t' - p^{-1} \mu_t \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_p' \\
&\quad - p^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \boldsymbol{\alpha}_t' - p^{-1} \mathbf{1}_p' \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}_p \mathbf{1}_p' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{C}_t' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' - q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' \\
&\quad - p^{-1} \mu_t \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' - p^{-1} \mu_t \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_p' - (pq)^{-1} p^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' + q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' \\
&\quad + q\mu_t \mathbf{1}_p \boldsymbol{\alpha}_t' + \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' + p^{-1} \mu_t \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \boldsymbol{\alpha}_t' + (pq)^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' \\
&\quad + q\mu_t^2 \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mu_t \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' + \boldsymbol{\beta}_t' \boldsymbol{\beta}_t \mathbf{1}_p \mathbf{1}_p' + p^{-1} \boldsymbol{\beta}_t' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mu_t \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_p' \\
&\quad + p^{-1} \mathbf{1}_p' \mathbf{E}_t \boldsymbol{\beta}_t \mathbf{1}_p \mathbf{1}_p' + p^{-2} \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p' \\
&= \mathbf{C}_t \mathbf{C}_t' + \mathbf{C}_t \mathbf{E}_t' + \mathbf{E}_t \mathbf{C}_t' + \mathbf{E}_t \mathbf{E}_t' + (pq)^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' + (pq)^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' \\
&\quad - p^{-1} \mathbf{C}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{C}_t' - p^{-1} \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' - q^{-1} \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \\
&\quad - (pq)^{-1} p^{-1} (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p)^2 \mathbf{1}_p \mathbf{1}_p' + p^{-2} \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p'.
\end{aligned} \tag{S.13}$$

Swapping the roles of row and column factor loadings, we can arrive at similarly

$$\begin{aligned}
\widehat{\mathbf{L}}_t' \widehat{\mathbf{L}}_t &= \mathbf{C}_t' \mathbf{C}_t + \mathbf{C}_t' \mathbf{E}_t + \mathbf{E}_t' \mathbf{C}_t + \mathbf{E}_t' \mathbf{E}_t + (pq)^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_q' + (pq)^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q \mathbf{1}_p' \mathbf{E}_t \\
&\quad - q^{-1} \mathbf{C}_t' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' - q^{-1} \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \mathbf{C}_t - q^{-1} \mathbf{E}_t' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' - q^{-1} \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \mathbf{E}_t - p^{-1} \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \\
&\quad - (pq)^{-1} q^{-1} (\mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q)^2 \mathbf{1}_q \mathbf{1}_q' + q^{-2} \mathbf{1}_q' \mathbf{E}_t' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q \mathbf{1}_q'.
\end{aligned} \tag{S.14}$$

For ease of notation, we define

$$\begin{aligned}
\mathbf{R}_{r,t} &:= \mathbf{C}_t \mathbf{E}_t' + \mathbf{E}_t \mathbf{C}_t' + \mathbf{E}_t \mathbf{E}_t' + (pq)^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' + (pq)^{-1} \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t' \\
&\quad - p^{-1} \mathbf{C}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{C}_t' - p^{-1} \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' - q^{-1} \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \\
&\quad - (pq)^{-1} p^{-1} (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p)^2 \mathbf{1}_p \mathbf{1}_p' + p^{-2} \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p',
\end{aligned} \tag{S.15}$$

$$\begin{aligned}
\mathbf{R}_{c,t} &:= \mathbf{C}_t' \mathbf{E}_t + \mathbf{E}_t' \mathbf{C}_t + \mathbf{E}_t' \mathbf{E}_t + (pq)^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_q' + (pq)^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q \mathbf{1}_p' \mathbf{E}_t' \\
&\quad - q^{-1} \mathbf{C}_t' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' - q^{-1} \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \mathbf{C}_t - q^{-1} \mathbf{E}_t' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' - q^{-1} \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \mathbf{E}_t - p^{-1} \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \\
&\quad - (pq)^{-1} q^{-1} (\mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q)^2 \mathbf{1}_q \mathbf{1}_q' + q^{-2} \mathbf{1}_q' \mathbf{E}_t' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q \mathbf{1}_q',
\end{aligned} \tag{S.16}$$

so that from (S.13) and (S.14), we can write

$$\widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t' = \mathbf{C}_t \mathbf{C}_t' + \mathbf{R}_{r,t}, \quad \widehat{\mathbf{L}}_t' \widehat{\mathbf{L}}_t = \mathbf{C}_t' \mathbf{C}_t + \mathbf{R}_{c,t}.$$

Recall that we denote by $\widehat{\mathbf{D}}_r$ the $k_r \times k_r$ diagonal matrix with the first largest k_r eigenvalues of $T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t'$ on the main diagonal, and since $\widehat{\mathbf{Q}}_r$ consists of the corresponding eigenvectors, we have

$$\widehat{\mathbf{Q}}_r \widehat{\mathbf{D}}_r = T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t' \widehat{\mathbf{Q}}_r. \tag{S.17}$$

With (S.13) and $\mathbf{C}_t \mathbf{C}_t' = \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' \mathbf{Q}_c \mathbf{F}_{Z,t}' \mathbf{Q}_r'$, we can write the j -th row of estimated row factor loading as

$$\begin{aligned}\widehat{\mathbf{Q}}_{r,j\cdot} &= T^{-1} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \widehat{\mathbf{Q}}_{r,i\cdot} \sum_{t=1}^T (\widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t')_{ij} \\ &= T^{-1} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \widehat{\mathbf{Q}}_{r,i\cdot} \mathbf{Q}_{r,i\cdot}' \sum_{t=1}^T (\mathbf{F}_{Z,t} \mathbf{Q}_c' \mathbf{Q}_c \mathbf{F}_{Z,t}') \mathbf{Q}_{r,j\cdot} + T^{-1} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \widehat{\mathbf{Q}}_{r,i\cdot} \sum_{t=1}^T (\mathbf{R}_{r,t})_{ij}.\end{aligned}$$

Thus with the definition $\mathbf{H}_r = T^{-1} \widehat{\mathbf{D}}_r^{-1} \widehat{\mathbf{Q}}_r' \mathbf{Q}_r \sum_{t=1}^T (\mathbf{F}_{Z,t} \mathbf{Q}_c' \mathbf{Q}_c \mathbf{F}_{Z,t}')$, we have

$$\widehat{\mathbf{Q}}_{r,j\cdot} - \mathbf{H}_r \mathbf{Q}_{r,j\cdot} = T^{-1} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \widehat{\mathbf{Q}}_{r,i\cdot} \sum_{t=1}^T (\mathbf{R}_{r,t})_{ij},$$

and hence we have

$$\begin{aligned}\|\widehat{\mathbf{Q}}_r - \mathbf{Q}_r \mathbf{H}_r'\|_F^2 &= \sum_{j=1}^p \|\widehat{\mathbf{Q}}_{r,j\cdot} - \mathbf{H}_r \mathbf{Q}_{r,j\cdot}\|^2 = \sum_{j=1}^p \left\| T^{-1} \widehat{\mathbf{D}}_r^{-1} \widehat{\mathbf{Q}}_r' \left(\sum_{t=1}^T \mathbf{R}_{r,t} \right)_{\cdot j} \right\|^2 \\ &\leq T^{-2} \cdot \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\widehat{\mathbf{Q}}_r\|_F^2 \cdot \left\| \sum_{t=1}^T \mathbf{R}_{r,t} \right\|_F^2 = O_P \left(T^{-1} p^{2(1-\delta_{r,k_r})} q^{1-2\delta_{c,1}} + p^{1-2\delta_{r,k_r}} q^{2(1-\delta_{c,1})} \right),\end{aligned}$$

where the last equality used Lemma 2 and Lemma 3. The consistency of $\widehat{\mathbf{Q}}_c$ can be similarly shown (omitted here). This completes the proof of Theorem 2. $\square$

Proof of Theorem 3. From (3.4), we can first write

$$\begin{aligned}\widehat{\mathbf{L}}_t &= \mathbf{Y}_t + (pq)^{-1} \mathbf{1}_p' \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_q' - q^{-1} \mathbf{Y}_t \mathbf{1}_q \mathbf{1}_q' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{Y}_t \\ &= \mu_t \mathbf{1}_p \mathbf{1}_q' + \alpha_t \mathbf{1}_q' + \mathbf{1}_p \beta_t' + \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' + \mathbf{E}_t \\ &\quad + (pq)^{-1} \mathbf{1}_p' (\mu_t \mathbf{1}_p \mathbf{1}_q' + \alpha_t \mathbf{1}_q' + \mathbf{1}_p \beta_t' + \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' + \mathbf{E}_t) \mathbf{1}_q \mathbf{1}_p \mathbf{1}_q' \\ &\quad - q^{-1} (\mu_t \mathbf{1}_p \mathbf{1}_q' + \alpha_t \mathbf{1}_q' + \mathbf{1}_p \beta_t' + \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' + \mathbf{E}_t) \mathbf{1}_q \mathbf{1}_q' \\ &\quad - p^{-1} \mathbf{1}_p \mathbf{1}_p' (\mu_t \mathbf{1}_p \mathbf{1}_q' + \alpha_t \mathbf{1}_q' + \mathbf{1}_p \beta_t' + \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' + \mathbf{E}_t) \\ &= \mu_t \mathbf{1}_p \mathbf{1}_q' + \alpha_t \mathbf{1}_q' + \mathbf{1}_p \beta_t' + \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' + \mathbf{E}_t + (pq)^{-1} \mathbf{1}_p' \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{1}_q \mathbf{1}_p \mathbf{1}_q' + (pq)^{-1} \mathbf{1}_p' \alpha_t \mathbf{1}_q' \mathbf{1}_q \mathbf{1}_p \mathbf{1}_q' \\ &\quad + (pq)^{-1} \mathbf{1}_p' \mathbf{1}_p \beta_t' \mathbf{1}_q \mathbf{1}_p \mathbf{1}_q' + (pq)^{-1} \mathbf{1}_p' \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' \mathbf{1}_q \mathbf{1}_p \mathbf{1}_q' + (pq)^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_q' \\ &\quad - q^{-1} \mu_t \mathbf{1}_p \mathbf{1}_q' \mathbf{1}_q \mathbf{1}_q' - q^{-1} \alpha_t \mathbf{1}_q' \mathbf{1}_q \mathbf{1}_q' - q^{-1} \mathbf{1}_p \beta_t' \mathbf{1}_q \mathbf{1}_q' - q^{-1} \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' \mathbf{1}_q \mathbf{1}_q' - q^{-1} \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \\ &\quad - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mu_t \mathbf{1}_p \mathbf{1}_q' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \alpha_t \mathbf{1}_q' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{1}_p \beta_t' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \\ &= \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}_c' + \mathbf{E}_t + (pq)^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p \mathbf{1}_q' - q^{-1} \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' - p^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t,\end{aligned}$$

where the last equality used Assumption (IC1). Thus, we have

$$\begin{aligned}
& \widehat{\mathbf{F}}_{Z,t} - (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} = \widehat{\mathbf{Q}}_r' \widehat{\mathbf{L}}_t \widehat{\mathbf{Q}}_c - (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} \\
& = \widehat{\mathbf{Q}}_r' (\mathbf{Q}_r \mathbf{H}_r') (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} (\mathbf{Q}_c \mathbf{H}_c')' \widehat{\mathbf{Q}}_c - (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} + \widehat{\mathbf{Q}}_r' \mathbf{E}_t \widehat{\mathbf{Q}}_c \\
& + (pq)^{-1} \widehat{\mathbf{Q}}_r' \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' \widehat{\mathbf{Q}}_c - q^{-1} \widehat{\mathbf{Q}}_r' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \widehat{\mathbf{Q}}_c - p^{-1} \widehat{\mathbf{Q}}_r' \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \widehat{\mathbf{Q}}_c \\
& = \widehat{\mathbf{Q}}_r' (\mathbf{Q}_r \mathbf{H}_r' - \widehat{\mathbf{Q}}_r) (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} (\mathbf{Q}_c \mathbf{H}_c' - \widehat{\mathbf{Q}}_c)' \widehat{\mathbf{Q}}_c + \widehat{\mathbf{Q}}_r' (\mathbf{Q}_r \mathbf{H}_r' - \widehat{\mathbf{Q}}_r) (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} \\
& + (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} (\mathbf{Q}_c \mathbf{H}_c' - \widehat{\mathbf{Q}}_c)' \widehat{\mathbf{Q}}_c + (\widehat{\mathbf{Q}}_r - \mathbf{Q}_r \mathbf{H}_r)' \mathbf{E}_t (\widehat{\mathbf{Q}}_c - \mathbf{Q}_c \mathbf{H}_c') \\
& + (\widehat{\mathbf{Q}}_r - \mathbf{Q}_r \mathbf{H}_r)' \mathbf{E}_t \mathbf{Q}_c \mathbf{H}_c' + \mathbf{H}_r' \mathbf{Q}_r' \mathbf{E}_t (\widehat{\mathbf{Q}}_c - \mathbf{Q}_c \mathbf{H}_c') + \mathbf{H}_r' \mathbf{Q}_r' \mathbf{E}_t \mathbf{Q}_c \mathbf{H}_c' \\
& + (pq)^{-1} \widehat{\mathbf{Q}}_r' \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' \widehat{\mathbf{Q}}_c - q^{-1} (\widehat{\mathbf{Q}}_r - \mathbf{Q}_r \mathbf{H}_r)' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \widehat{\mathbf{Q}}_c - q^{-1} \mathbf{H}_r \mathbf{Q}_r' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \widehat{\mathbf{Q}}_c \\
& - p^{-1} \widehat{\mathbf{Q}}_r' \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t (\widehat{\mathbf{Q}}_c - \mathbf{Q}_c \mathbf{H}_c') - p^{-1} \widehat{\mathbf{Q}}_r' \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{Q}_c \mathbf{H}_c' \\
& =: \mathcal{I}_{F,1} + \mathcal{I}_{F,2} + \mathcal{I}_{F,3} + \mathcal{I}_{F,4} + \mathcal{I}_{F,5} + \mathcal{I}_{F,6} + \mathcal{I}_{F,7} + \mathcal{I}_{F,8} \\
& - \mathcal{I}_{F,9} - \mathcal{I}_{F,10} - \mathcal{I}_{F,11} - \mathcal{I}_{F,12}, \quad \text{where}
\end{aligned} \tag{S.18}$$

$$\begin{aligned}
\mathcal{I}_{F,1} &:= \widehat{\mathbf{Q}}_r' (\mathbf{Q}_r \mathbf{H}_r' - \widehat{\mathbf{Q}}_r) (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} (\mathbf{Q}_c \mathbf{H}_c' - \widehat{\mathbf{Q}}_c)' \widehat{\mathbf{Q}}_c, \\
\mathcal{I}_{F,2} &:= \widehat{\mathbf{Q}}_r' (\mathbf{Q}_r \mathbf{H}_r' - \widehat{\mathbf{Q}}_r) (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1}, \quad \mathcal{I}_{F,3} := (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} (\mathbf{Q}_c \mathbf{H}_c' - \widehat{\mathbf{Q}}_c)' \widehat{\mathbf{Q}}_c, \\
\mathcal{I}_{F,4} &:= (\widehat{\mathbf{Q}}_r - \mathbf{Q}_r \mathbf{H}_r)' \mathbf{E}_t (\widehat{\mathbf{Q}}_c - \mathbf{Q}_c \mathbf{H}_c'), \quad \mathcal{I}_{F,5} := (\widehat{\mathbf{Q}}_r - \mathbf{Q}_r \mathbf{H}_r)' \mathbf{E}_t \mathbf{Q}_c \mathbf{H}_c', \\
\mathcal{I}_{F,6} &:= \mathbf{H}_r' \mathbf{Q}_r' \mathbf{E}_t (\widehat{\mathbf{Q}}_c - \mathbf{Q}_c \mathbf{H}_c'), \quad \mathcal{I}_{F,7} := \mathbf{H}_r' \mathbf{Q}_r' \mathbf{E}_t \mathbf{Q}_c \mathbf{H}_c', \quad \mathcal{I}_{F,8} := (pq)^{-1} \widehat{\mathbf{Q}}_r' \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' \widehat{\mathbf{Q}}_c, \\
\mathcal{I}_{F,9} &:= q^{-1} (\widehat{\mathbf{Q}}_r - \mathbf{Q}_r \mathbf{H}_r)' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \widehat{\mathbf{Q}}_c, \quad \mathcal{I}_{F,10} := q^{-1} \mathbf{H}_r \mathbf{Q}_r' \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \widehat{\mathbf{Q}}_c, \\
\mathcal{I}_{F,11} &:= p^{-1} \widehat{\mathbf{Q}}_r' \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t (\widehat{\mathbf{Q}}_c - \mathbf{Q}_c \mathbf{H}_c'), \quad \mathcal{I}_{F,12} := p^{-1} \widehat{\mathbf{Q}}_r' \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{Q}_c \mathbf{H}_c'.
\end{aligned}$$

First consider $\mathbf{F}_{Z,t}$, by its definition and Assumption (L1) we have

$$\|\mathbf{F}_{Z,t}\|_F^2 \leq \|\mathbf{F}_t\|_F^2 \cdot \|\mathbf{Z}_r^{1/2}\|_F^2 \cdot \|\mathbf{Z}_c^{1/2}\|_F^2 = O_P(p^{\delta_{r,1}} q^{\delta_{c,1}}).$$

Then for $\mathcal{I}_{F,1}$, we have

$$\begin{aligned}
\|\mathcal{I}_{F,1}\|_F^2 &= O_P(p^{\delta_{r,1}} q^{\delta_{c,1}}) \cdot \|\mathbf{Q}_r \mathbf{H}_r' - \widehat{\mathbf{Q}}_r\|_F^2 \cdot \|\mathbf{Q}_c \mathbf{H}_c' - \widehat{\mathbf{Q}}_c\|_F^2 \\
&= O_P(T^{-2} p^{2-2\delta_{r,k_r}} q^{2-2\delta_{c,k_c}} + T^{-1} p^{3-2\delta_{r,k_r}} q^{2-\delta_{c,1}-3\delta_{c,k_c}} \\
&\quad + T^{-1} q^{3-2\delta_{c,k_c}} p^{2-\delta_{r,1}-3\delta_{r,k_r}} + p^{3-\delta_{r,1}-3\delta_{r,k_r}} q^{3-\delta_{c,1}-3\delta_{c,k_c}}),
\end{aligned}$$

where we used Lemma 6 in the last equality. Similarly, we can obtain the following for $\mathcal{I}_{F,2}$ and $\mathcal{I}_{F,3}$,

$$\begin{aligned}
\|\mathcal{I}_{F,2}\|_F^2 &= O_P(T^{-1} p^{1+2\delta_{r,1}-2\delta_{r,k_r}} q^{1-\delta_{c,1}} + p^{1+\delta_{r,1}-3\delta_{r,k_r}} q^{2-\delta_{c,1}}), \\
\|\mathcal{I}_{F,3}\|_F^2 &= O_P(T^{-1} q^{1+2\delta_{c,1}-2\delta_{c,k_c}} p^{1-\delta_{r,1}} + q^{1+\delta_{c,1}-3\delta_{c,k_c}} p^{2-\delta_{r,1}}).
\end{aligned}$$

For $\mathcal{I}_{F,4}$, from Assumptions (E1) and (E2) we easily have $\|\mathbf{E}_t\|_F^2 = O(pq)$, so we have

$$\begin{aligned}
\|\mathcal{I}_{F,4}\|_F^2 &= O_P(pq) \cdot \|\mathbf{Q}_r \mathbf{H}_r' - \widehat{\mathbf{Q}}_r\|_F^2 \cdot \|\mathbf{Q}_c \mathbf{H}_c' - \widehat{\mathbf{Q}}_c\|_F^2 \\
&= O_P(T^{-2} p^{3-\delta_{r,1}-2\delta_{r,k_r}} q^{3-\delta_{c,1}-2\delta_{c,k_c}} + T^{-1} p^{4-\delta_{r,1}-2\delta_{r,k_r}} q^{3-2\delta_{c,1}-3\delta_{c,k_c}} \\
&\quad + T^{-1} q^{4-\delta_{c,1}-2\delta_{c,k_c}} p^{3-2\delta_{r,1}-3\delta_{r,k_r}} + p^{4-2\delta_{r,1}-3\delta_{r,k_r}} q^{4-2\delta_{c,1}-3\delta_{c,k_c}}),
\end{aligned}$$

For $\mathcal{I}_{F,5}$, consider first

$$\begin{aligned}
& \mathbb{E} \left\{ \|\mathbf{E}_t \mathbf{A}_c\|_F^2 \right\} \leq pk_c \max_{i \in [p], j \in [k_c]} \mathbb{E} \left\{ (\mathbf{E}_{t,i}' \mathbf{A}_{c,j})^2 \right\} \\
& = pk_c \max_{i \in [p], j \in [k_c]} \sum_{n=1}^q \sum_{l=1}^q \text{cov}(E_{t,in}, E_{t,il}) \cdot A_{c,nj} A_{c,lj} = O(pq),
\end{aligned} \tag{S.19}$$

where the last equality used Assumptions (E1) and (E2). Thus,

$$\begin{aligned}\|\mathcal{I}_{F,5}\|_F^2 &= O_P(pq) \cdot \|\mathbf{Q}_r \mathbf{H}'_r - \widehat{\mathbf{Q}}_r\|_F^2 \cdot \|\mathbf{Z}_c^{-1/2}\|_F^2 \\ &= O_P(T^{-1} p^{2+\delta_{r,1}-2\delta_{r,k_r}} q^{2-2\delta_{c,1}-\delta_{c,k_c}} + p^{2-3\delta_{r,k_r}} q^{3-2\delta_{c,1}-\delta_{c,k_c}}),\end{aligned}$$

where we used Lemma 6 and Assumption (L1). Similarly for $\mathcal{I}_{F,6}$,

$$\|\mathcal{I}_{F,6}\|_F^2 = O_P(T^{-1} q^{2+\delta_{c,1}-2\delta_{c,k_c}} p^{2-2\delta_{r,1}-\delta_{r,k_r}} + q^{2-3\delta_{c,k_c}} p^{3-2\delta_{r,1}-\delta_{r,k_r}}).$$

By Assumptions (E1) and (E2) again, we have

$$\begin{aligned}\mathbb{E}\left\{\|\mathbf{A}'_r \mathbf{E}_t \mathbf{A}_c\|_F^2\right\} &\leq k_r k_c \max_{i \in [k_r], j \in [k_c]} \mathbb{E}\left\{(\mathbf{A}'_{r,i} \mathbf{E}_t \mathbf{A}_{c,j})^2\right\} \\ &= k_r k_c \max_{i \in [k_r], j \in [k_c]} \sum_{m=1}^p \sum_{n=1}^q \sum_{h=1}^p \sum_{l=1}^q \text{cov}(E_{t,mn}, E_{t,hl}) \cdot A_{r,mi} A_{c,nj} A_{r,hi} A_{c,lj} = O(pq),\end{aligned}$$

hence for $\mathcal{I}_{F,7}$, it holds that

$$\|\mathcal{I}_{F,7}\|_F^2 = O_P(pq) \cdot \|\mathbf{Z}_r^{-1/2}\|_F^2 \cdot \|\mathbf{Z}_c^{-1/2}\|_F^2 = O_P(p^{1-\delta_{r,k_r}} q^{1-\delta_{c,k_c}}).$$

Consider $\mathcal{I}_{F,8}$, recall that $(\mathbf{1}'_p \mathbf{E}_t \mathbf{1}_q)^2 = O_P(pq)$ from (S.2) and hence

$$\|\mathcal{I}_{F,8}\|_F^2 = O_P(p^{-2} q^{-2}) \cdot (\mathbf{1}'_p \mathbf{E}_t \mathbf{1}_q)^2 \cdot \|\mathbf{1}_p \mathbf{1}'_q\|_F^2 = O_P(1).$$

For $\mathcal{I}_{F,9}$, note that $\mathbb{E}[\|\mathbf{E}_t \mathbf{1}_q\|_F^2]$ has the same rate as (S.19) since k_c is fixed, and hence

$$\begin{aligned}\|\mathcal{I}_{F,9}\|_F^2 &= O_P(q^{-2}) \cdot \|\mathbf{Q}_r \mathbf{H}'_r - \widehat{\mathbf{Q}}_r\|_F^2 \cdot \|\mathbf{E}_t \mathbf{1}_q\|_F^2 \cdot \|\mathbf{1}_q\|^2 \\ &= O_P(T^{-1} p^{2+\delta_{r,1}-2\delta_{r,k_r}} q^{1-2\delta_{c,1}} + p^{2-3\delta_{r,k_r}} q^{2-2\delta_{c,1}}).\end{aligned}$$

From (S.2), we also have $\mathbb{E}[\|\mathbf{A}'_r \mathbf{E}_t \mathbf{1}_q\|_F^2] = O(pq)$, so for $\mathcal{I}_{F,10}$ we have

$$\|\mathcal{I}_{F,10}\|_F^2 = O_P(q^{-2}) \cdot \|\mathbf{Z}_r^{-1/2}\|_F^2 \cdot \|\mathbf{A}'_r \mathbf{E}_t \mathbf{1}_q\|_F^2 \cdot \|\mathbf{1}_q\|^2 = O_P(p^{1-\delta_{r,k_r}}).$$

Lastly, the rates for $\mathcal{I}_{F,11}$ and $\mathcal{I}_{F,12}$ can be obtained similarly as $\mathcal{I}_{F,9}$ and $\mathcal{I}_{F,10}$,

$$\begin{aligned}\|\mathcal{I}_{F,11}\|_F^2 &= O_P(T^{-1} q^{2+\delta_{c,1}-2\delta_{c,k_c}} p^{1-2\delta_{r,1}} + q^{2-3\delta_{c,k_c}} p^{2-2\delta_{r,1}}), \\ \|\mathcal{I}_{F,12}\|_F^2 &= O_P(q^{1-\delta_{c,k_c}}).\end{aligned}$$

Therefore, with all the rates from $\mathcal{I}_{F,1}$ to $\mathcal{I}_{F,12}$ in (S.18), by using Assumption (R1) we have

$$\begin{aligned}\|\widehat{\mathbf{F}}_{Z,t} - (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1}\|_F^2 &= O_P(p^{1-\delta_{r,k_r}} q^{1-\delta_{c,k_c}} + T^{-1} p^{1+2\delta_{r,1}-2\delta_{r,k_r}} q^{1-\delta_{c,1}} + p^{1+\delta_{r,1}-3\delta_{r,k_r}} q^{2-\delta_{c,1}} \\ &\quad + T^{-1} q^{1+2\delta_{c,1}-2\delta_{c,k_c}} p^{1-\delta_{r,1}} + q^{1+\delta_{c,1}-3\delta_{c,k_c}} p^{2-\delta_{r,1}}).\end{aligned}$$

This shows the first statement of Theorem 3.

For the remaining proof, consider $\widehat{C}_{t,ij} - C_{t,ij}$ for any $t \in [T], i \in [p], j \in [q]$. First, we have

$$\begin{aligned}\widehat{C}_{t,ij} - C_{t,ij} &= \widehat{\mathbf{Q}}'_{r,i} \widehat{\mathbf{F}}_{Z,t} \widehat{\mathbf{Q}}_{c,j} - \mathbf{Q}'_{r,i} \mathbf{F}_{Z,t} \mathbf{Q}_{c,j} \\ &= (\widehat{\mathbf{Q}}_{r,i} - \mathbf{H}_r \mathbf{Q}_{r,i})' (\widehat{\mathbf{F}}_{Z,t} - (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1}) (\widehat{\mathbf{Q}}_{c,j} - \mathbf{H}_c \mathbf{Q}_{c,j}) \\ &\quad + (\widehat{\mathbf{Q}}_{r,i} - \mathbf{H}_r \mathbf{Q}_{r,i})' (\widehat{\mathbf{F}}_{Z,t} - (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1}) \mathbf{H}_c \mathbf{Q}_{c,j} + (\widehat{\mathbf{Q}}_{r,i} - \mathbf{H}_r \mathbf{Q}_{r,i})' (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} (\widehat{\mathbf{Q}}_{c,j} - \mathbf{H}_c \mathbf{Q}_{c,j}) \\ &\quad + (\widehat{\mathbf{Q}}_{r,i} - \mathbf{H}_r \mathbf{Q}_{r,i})' (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{Q}_{c,j} + \mathbf{Q}'_{r,i} \mathbf{H}_r' (\widehat{\mathbf{F}}_{Z,t} - (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1}) (\widehat{\mathbf{Q}}_{c,j} - \mathbf{H}_c \mathbf{Q}_{c,j}) \\ &\quad + \mathbf{Q}'_{r,i} \mathbf{H}_r' (\widehat{\mathbf{F}}_{Z,t} - (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1}) \mathbf{H}_c \mathbf{Q}_{c,j} + \mathbf{Q}'_{r,i} \mathbf{F}_{Z,t} \mathbf{H}_c^{-1} (\widehat{\mathbf{Q}}_{c,j} - \mathbf{H}_c \mathbf{Q}_{c,j}).\end{aligned}\tag{S.20}$$

Notice that from Assumption (L1),

$$\begin{aligned}\|\mathbf{Q}_{r,i}\|^2 &= O(\|\mathbf{A}_{r,i}\|^2) \cdot \|\mathbf{Z}_r^{-1/2}\|_F^2 = O(p^{-\delta_{r,k_r}}), \\ \|\mathbf{Q}_{c,j}\|^2 &= O(\|\mathbf{A}_{c,j}\|^2) \cdot \|\mathbf{Z}_c^{-1/2}\|_F^2 = O(q^{-\delta_{c,k_c}}).\end{aligned}$$

Together with Lemma 6, the first statement of Theorem 3 and Assumption (R1), we have

$$\begin{aligned}(\hat{C}_{t,ij} - C_{t,ij})^2 &= O_P(p^{1-2\delta_{r,k_r}} q^{1-2\delta_{c,k_c}} + T^{-1} p^{1+2\delta_{r,1}-3\delta_{r,k_r}} q^{1-\delta_{c,1}-\delta_{c,k_c}} + p^{1+\delta_{r,1}-4\delta_{r,k_r}} q^{2-\delta_{c,1}-\delta_{c,k_c}} \\ &\quad + T^{-1} q^{1+2\delta_{c,1}-3\delta_{c,k_c}} p^{1-\delta_{r,1}-\delta_{r,k_r}} + q^{1+\delta_{c,1}-4\delta_{c,k_c}} p^{2-\delta_{r,1}-\delta_{r,k_r}}).\end{aligned}$$

This completes the proof of Theorem 3. $\square$

Proof of Theorem 4. We first consider $\hat{\mu}_t$, which is given by

$$\begin{aligned}\hat{\mu}_t &= \mathbf{1}'_p \mathbf{Y}_t \mathbf{1}_q / (pq) = \mu_t + \mathbf{1}'_p \mathbf{E}_t \mathbf{1}_q / (pq) = \mu_t + \frac{1}{pq} \sum_{i,j} E_{t,ij}, \text{ so that} \\ \hat{\mu}_t - \mu_t &= \frac{1}{pq} \mathbf{1}'_p \mathbf{A}_{e,r} \mathbf{F}_{e,t} \mathbf{A}'_{e,c} \mathbf{1}_q + \frac{1}{pq} \sum_{i,j} \Sigma_{\epsilon,ij} \epsilon_{t,ij} =: I_{\mu,1} + I_{\mu,2}.\end{aligned}$$

By Assumption (E1), since $\|\mathbf{A}_{e,r}\|_1, \|\mathbf{A}_{e,c}\|_1 = O(1)$, we have $I_{\mu,1} = O_P(1/(pq))$. Also, $I_{\mu,2} = O((pq)^{-1/2})$ since $\mathbb{E}(I_{\mu,2}) = 0$ and $\text{Var}(I_{\mu,2}) = (pq)^{-2} \sum_{i,j} \Sigma_{\epsilon,ij}^2 = O(1/(pq))$. Hence $I_{\mu,1}$ is dominated by $I_{\mu,2}$, and

$$\sqrt{pq}(\hat{\mu}_t - \mu_t) = \sqrt{pq} I_{\mu,2} (1 + o_P(1)) = \frac{1}{\sqrt{pq}} \sum_{i,j} \Sigma_{\epsilon,ij} \epsilon_{t,ij} (1 + o_P(1)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \gamma_\mu^2),$$

where we use Theorem 1 in Ayvazyan and Ulyanov (2023) for the convergence in distribution.

For $\hat{\alpha}_t$, consider the decomposition

$$\begin{aligned}\hat{\alpha}_t - \alpha_t &= (\mu_t - \hat{\mu}_t) \mathbf{1}_p + q^{-1} \mathbf{E}_t \mathbf{1}_q \\ &= (\mu_t - \hat{\mu}_t) \mathbf{1}_p + q^{-1} \mathbf{A}_{e,r} \mathbf{F}_{e,t} \mathbf{A}'_{e,c} \mathbf{1}_q + q^{-1} (\Sigma_\epsilon \circ \epsilon_t) \mathbf{1}_q, \text{ so that} \\ \hat{\alpha}_{t,i} - \alpha_{t,i} &= (\mu_t - \hat{\mu}_t) + q^{-1} (\mathbf{A}_{e,r})_i \cdot \mathbf{F}_{e,t} \mathbf{A}'_{e,c} \mathbf{1}_q + q^{-1} \sum_{j=1}^q \Sigma_{\epsilon,ij} \epsilon_{t,ij}.\end{aligned}$$

From what we have proved above we have $|\hat{\mu}_t - \mu_t| = O_P(1/\sqrt{pq})$. Also, $q^{-1} (\mathbf{A}_{e,r})_i \cdot \mathbf{F}_{e,t} \mathbf{A}'_{e,c} \mathbf{1}_q = O_P(1/q)$ since $\|\mathbf{A}_{e,r}\|_1, \|\mathbf{A}_{e,c}\|_1 = O(1)$. Finally, $\mathbb{E}(q^{-1} (\Sigma_\epsilon \circ \epsilon_t) \mathbf{1}_q) = \mathbf{0}$ and $\text{Var}(q^{-1} \sum_{j=1}^q \Sigma_{\epsilon,ij} \epsilon_{t,ij}) = q^{-2} \sum_{j=1}^q \Sigma_{\epsilon,ij}^2 = O(q^{-1})$, implying that $q^{-1} (\Sigma_\epsilon \circ \epsilon_t) \mathbf{1}_q = O_P(q^{-1/2})$ elementwise, thus dominating other terms. Hence for $i \in [p]$, using Theorem 1 in Ayvazyan and Ulyanov (2023),

$$\sqrt{q}(\hat{\alpha}_{t,i} - \alpha_{t,i}) = q^{-1/2} \sum_{j=1}^q \Sigma_{\epsilon,ij} \epsilon_{t,ij} (1 + o_P(1)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \gamma_{\alpha,i}^2).$$

By Assumption (E1), each element of $q^{-1} (\Sigma_\epsilon \circ \epsilon_t) \mathbf{1}_q$ is independent of each other. Hence with integers $i_1 < \dots < i_m$, m being finite and $\boldsymbol{\theta}_{\alpha,t} := (\alpha_{t,i_1}, \dots, \alpha_{t,i_m})'$, by Theorem 1 in Ayvazyan and Ulyanov (2023),

$$\sqrt{q}(\hat{\boldsymbol{\theta}}_{\alpha,t} - \boldsymbol{\theta}_{\alpha,t}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \text{diag}(\gamma_{\alpha,i_1}^2, \dots, \gamma_{\alpha,i_m}^2)).$$

We omit the proof of asymptotic normality for $\hat{\beta}_t$ since the arguments used are in parallel to those used for $\hat{\alpha}_t$, using the independence of the columns in $\Sigma_\epsilon \circ \epsilon_t$ by Assumption (E1).

The rest of the proof is done if we can prove that $\hat{\gamma}_{\alpha,i}$, $\hat{\gamma}_{\beta,j}$ and $\hat{\gamma}_\mu$ are consistent estimators for $\gamma_{\alpha,i}$, $\gamma_{\beta,j}$ and γ_μ respectively. From (3.6), since we assume $\hat{C}_{t,ij} - C_{t,ij} = o_P(1)$ from Theorem 3, then elementwise we have

$$\hat{\mathbf{E}}_t = \hat{\mathbf{L}}_t - \hat{\mathbf{C}}_t = (\mu_t - \hat{\mu}_t) \mathbf{1}_p \mathbf{1}'_q + (\alpha_t - \hat{\alpha}_t) \mathbf{1}'_q + \mathbf{1}_p (\beta_t - \hat{\beta}_t)' + (\mathbf{C}_t - \hat{\mathbf{C}}_t) + \mathbf{E}_t = \mathbf{E}_t (1 + o_P(1)).$$

Hence we have

$$\begin{aligned}
q^{-1}(\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t')_{ii} &= \{q^{-1}(\mathbf{A}_{e,r})_i \mathbf{F}_{e,t} \mathbf{A}'_{e,c} \mathbf{A}_{e,c} \mathbf{F}'_{e,t} (\mathbf{A}_{e,r})'_i + q^{-1}(\mathbf{A}_{e,r})_i \mathbf{F}_{e,t} \mathbf{A}'_{e,c} (\boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\epsilon}_t)'_i \\
&\quad + q^{-1}(\boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\epsilon}_t)_i (\boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\epsilon}_t)'_i\} (1 + o_P(1)) \\
&= O_P(q^{-1}) + O_P(q^{-1}) + \frac{1}{q} \sum_{j=1}^q \Sigma_{\epsilon,ij}^2 (\epsilon_{t,ij}^2 - 1) (1 + o_P(1)) + \frac{1}{q} \sum_{j=1}^q \Sigma_{\epsilon,ij}^2 \\
&\xrightarrow{P} \gamma_{\alpha,i}^2,
\end{aligned}$$

where we used the Markov inequality to arrive at $q^{-1} \sum_{j=1}^q \Sigma_{\epsilon,ij}^2 (\epsilon_{t,ij}^2 - 1) = O_P(q^{-1/2})$, knowing that each $\Sigma_{\epsilon,ij}$ is bounded away from infinity by Assumption (E1). A parallel argument (omitted) can show that $\widehat{\gamma}_{\beta,j}^2$ is consistent for $\gamma_{\beta,j}^2$. Finally,

$$\widehat{\gamma}_\mu^2 = p^{-1} \sum_{i=1}^p q^{-1}(\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t')_{ii} \xrightarrow{P} p^{-1} \sum_{i=1}^p \gamma_{\alpha,i}^2 = \gamma_\mu^2.$$

This completes the proof of the theorem. $\square$

Proof of Theorem 5. We construct the asymptotic normality for rows of our factor loading estimators. We only prove the result for the row loading estimator, and the proof for the column loading estimator would be similar (and hence omitted). For any $j \in [p]$, consider the following decomposition

$$\begin{aligned}
\widehat{\mathbf{Q}}_{r,j} - \mathbf{H}_r \mathbf{Q}_{r,j} &= T^{-1} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \widehat{\mathbf{Q}}_{r,i} \cdot \sum_{t=1}^T (\mathbf{R}_{r,t})_{ij} \\
&= T^{-1} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p (\widehat{\mathbf{Q}}_{r,i} - \mathbf{H}_r \mathbf{Q}_{r,i}) \sum_{t=1}^T (\mathbf{R}_{r,t})_{ij} + T^{-1} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{R}_{r,t})_{ij} \quad (\text{S.21}) \\
&= \mathcal{I}_1 + \mathcal{I}_2 + \mathcal{I}_3 + \mathcal{I}_4 + \mathcal{I}_5 - \mathcal{I}_6 - \mathcal{I}_7 - \mathcal{I}_8 - \mathcal{I}_9 - \mathcal{I}_{10} - \mathcal{I}_{11} + \mathcal{I}_{12} + \mathcal{I}_{13}, \quad \text{where} \\
\mathcal{I}_1 &:= \frac{1}{T} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}_t')_{ij}, \quad \mathcal{I}_2 := \frac{1}{T} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{E}_t \mathbf{C}_t')_{ij}, \\
\mathcal{I}_3 &:= \frac{1}{T} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{E}_t \mathbf{E}_t')_{ij}, \quad \mathcal{I}_4 := \frac{1}{T p q} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p')_{ij}, \\
\mathcal{I}_5 &:= \frac{1}{T p q} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t')_{ij}, \quad \mathcal{I}_6 := \frac{1}{T p} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p')_{ij}, \\
\mathcal{I}_7 &:= \frac{1}{T p} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{C}_t')_{ij}, \quad \mathcal{I}_8 := \frac{1}{T p} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p')_{ij}, \\
\mathcal{I}_9 &:= \frac{1}{T p} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t')_{ij}, \quad \mathcal{I}_{10} := \frac{1}{T p} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t')_{ij}, \\
\mathcal{I}_{11} &:= \frac{1}{T p^2 q} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T ((\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p)^2 \mathbf{1}_p \mathbf{1}_p')_{ij}, \quad \mathcal{I}_{12} := \frac{1}{T p^2} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p')_{ij}, \\
\mathcal{I}_{13} &:= \frac{1}{T} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p (\widehat{\mathbf{Q}}_{r,i} - \mathbf{H}_r \mathbf{Q}_{r,i}) \sum_{t=1}^T (\mathbf{R}_{r,t})_{ij}.
\end{aligned}$$

We shall show that $\mathcal{I}_1$ is the leading term among the decomposition in (S.21). To obtain the rate for $\mathcal{I}_2$, from Assumptions (E1) and (E2) we have for any $i \in [p], h \in [q]$,

$$E_{t,ih} = \sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,r,i} \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h} + (\boldsymbol{\Sigma}_\epsilon)_{ih} \sum_{w \geq 0} a_{e,w} (\mathbf{X}_{e,t-w})_{ih}.$$

Consider first $\sum_{t=1}^T \sum_{h=1}^q (\sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,r,i} \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h}) \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,j}$. By Assumptions (F1), (E1) and (E2), we have

$$\begin{aligned}
& \mathbb{E} \left\{ \left(\sum_{t=1}^T \sum_{h=1}^q \left(\sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,r,i} \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h} \right) \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,j} \right)^2 \right\} \\
&= \text{cov} \left(\sum_{t=1}^T \sum_{h=1}^q \mathbf{A}'_{c,h} \left(\sum_{w \geq 0} a_{f,w} \mathbf{X}'_{f,t-w} \right) \mathbf{A}_{r,j}, \left(\sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,r,i} \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h} \right), \right. \\
&\quad \left. \sum_{t=1}^T \sum_{h=1}^q \mathbf{A}'_{c,h} \left(\sum_{w \geq 0} a_{f,w} \mathbf{X}'_{f,t-w} \right) \mathbf{A}_{r,j}, \left(\sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,r,i} \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h} \right) \right) \\
&= \sum_{h=1}^q \sum_{l=1}^q \sum_{t=1}^T \sum_{w \geq 0} a_{f,w}^2 a_{e,w}^2 \cdot \|\mathbf{A}_{r,j}\|^2 \cdot \|\mathbf{A}_{c,h}\| \cdot \|\mathbf{A}_{c,l}\| \cdot \|\mathbf{A}_{e,c,h}\| \cdot \|\mathbf{A}_{e,c,l}\| \cdot \|\mathbf{A}_{e,r,i}\|^2 \\
&= O(T) \cdot \|\mathbf{A}_{r,j}\|^2 \cdot \|\mathbf{A}_{e,r,i}\|^2.
\end{aligned} \tag{S.22}$$

Consider also $\sum_{i=1}^p \sum_{h=1}^q \sum_{t=1}^T \mathbf{Q}_{r,i} ((\boldsymbol{\Sigma}_\epsilon)_{ih} \sum_{w \geq 0} a_{\epsilon,w} (\mathbf{X}_{\epsilon,t-w})_{ih}) \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,j}$. Similarly, by Assumptions (E1), (E2) and (F1), we have

$$\begin{aligned}
& \mathbb{E} \left\{ \left\| \sum_{i=1}^p \sum_{h=1}^q \sum_{t=1}^T \mathbf{Q}_{r,i} ((\boldsymbol{\Sigma}_\epsilon)_{ih} \sum_{w \geq 0} a_{\epsilon,w} (\mathbf{X}_{\epsilon,t-w})_{ih}) \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,j} \right\|^2 \right\} \\
&= \text{cov} \left(\sum_{i=1}^p \sum_{h=1}^q \sum_{t=1}^T \mathbf{Q}_{r,i} ((\boldsymbol{\Sigma}_\epsilon)_{ih} \sum_{w \geq 0} a_{\epsilon,w} (\mathbf{X}_{\epsilon,t-w})_{ih}) \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,j}, \right. \\
&\quad \left. \sum_{i=1}^p \sum_{h=1}^q \sum_{t=1}^T \mathbf{Q}_{r,i} ((\boldsymbol{\Sigma}_\epsilon)_{ih} \sum_{w \geq 0} a_{\epsilon,w} (\mathbf{X}_{\epsilon,t-w})_{ih}) \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,j} \right) \\
&= \sum_{i=1}^p \sum_{h=1}^q \sum_{t=1}^T \sum_{w \geq 0} a_{f,w}^2 a_{\epsilon,w}^2 \cdot \|\mathbf{A}_{r,j}\|^2 \cdot \|\mathbf{A}_{c,h}\|^2 \cdot (\boldsymbol{\Sigma}_\epsilon)_{ih}^2 \cdot \|\mathbf{Q}_{r,i}\|^2 \\
&= O(T) \cdot \|\mathbf{A}_{r,j}\|^2 \cdot \|\mathbf{A}_c\|^2 \cdot \|\mathbf{Q}_r\|^2.
\end{aligned} \tag{S.23}$$

Hence using Lemma 3, it holds that

$$\begin{aligned}
\|\mathcal{I}_2\|^2 &\leq \frac{1}{T^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \sum_{t=1}^T \sum_{h=1}^q E_{t,ih} \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,j} \right\|^2 \\
&= O_P \left(\frac{1}{T^2} p^{-2\delta_{r,k_r}} q^{-2\delta_{c,1}} \right) \left\{ \left\| \sum_{i=1}^p \sum_{h=1}^q \sum_{t=1}^T \mathbf{Q}_{r,i} ((\boldsymbol{\Sigma}_\epsilon)_{ih} \sum_{w \geq 0} a_{\epsilon,w} (\mathbf{X}_{\epsilon,t-w})_{ih}) \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,j} \right\|^2 \right. \\
&\quad \left. + \left(\sum_{i=1}^p \|\mathbf{Q}_{r,i}\|^2 \right) \sum_{i=1}^p \left(\sum_{t=1}^T \sum_{h=1}^q \left(\sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,r,i} \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h} \right) \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,j} \right)^2 \right\} \\
&= O_P \left(\frac{1}{T^2} p^{-2\delta_{r,k_r}} q^{-2\delta_{c,1}} \cdot T q^{\delta_{c,1}} \right) = O_P \left(T^{-1} p^{-2\delta_{r,k_r}} q^{-\delta_{c,1}} \right),
\end{aligned}$$

where we used Assumption (L1), (S.22) and (S.23) in the last equality.

For $\mathcal{I}_3$, first notice from the noise structure in Assumptions (E1) and (E2), we have

$$\begin{aligned}
& \text{Var}\left(\sum_{i=1}^p \sum_{h=1}^q \sum_{t=1}^T \mathbf{Q}_{r,i} E_{t,ih} E_{t,jh}\right) \\
&= O(1) \cdot \sum_{i=1}^p \sum_{u=1}^p \sum_{h=1}^q \sum_{l=1}^q \sum_{t=1}^T \sum_{n=1}^{k_{e,r}} \sum_{m=1}^{k_{e,c}} \sum_{w \geq 0} a_{e,w}^4 A_{e,r,in} A_{e,r,un} A_{e,r,jn}^2 A_{e,c,hm}^2 A_{e,c,lm}^2 \\
&\quad \cdot \|\mathbf{Q}_{r,i}\| \cdot \|\mathbf{Q}_{r,u}\| \cdot \text{Var}((\mathbf{X}_{e,t-w})_{nm}^2) \\
&\quad + O(1) \cdot \sum_{i=1}^p \sum_{h=1}^q \sum_{t=1}^T \sum_{w \geq 0} a_{e,w}^4 (\boldsymbol{\Sigma}_\epsilon)_{ih}^2 (\boldsymbol{\Sigma}_\epsilon)_{jh}^2 \cdot \|\mathbf{Q}_{r,i}\|^2 \cdot \text{Var}((\mathbf{X}_{e,t-w})_{ih} (\mathbf{X}_{e,t-w})_{jh}) \\
&= O(T + Tq) = O(Tq).
\end{aligned}$$

Moreover, it holds that

$$\mathbb{E}\left(\sum_{i=1}^p \left| \sum_{h=1}^q \sum_{t=1}^T E_{t,ih} E_{t,jh} \right| \right) = \sum_{i=1}^p \left| \sum_{h=1}^q \sum_{t=1}^T (\|\mathbf{A}_{e,c,h}\|^2 \cdot \|\mathbf{A}_{e,r,i}\| \cdot \|\mathbf{A}_{e,r,j}\| + (\boldsymbol{\Sigma}_\epsilon)_{ih} \mathbb{1}_{\{i=j\}}) \right| = O(Tq),$$

and with $\max_i \|\mathbf{Q}_{r,i}\|^2 \leq \|\mathbf{A}_{r,j}\|^2 \cdot \|\mathbf{Z}_r^{-1/2}\|^2 = O_P(p^{-\delta_r, k_r})$, we thus have

$$\begin{aligned}
\|\mathcal{I}_3\|^2 &\leq \frac{1}{T^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T \sum_{h=1}^q E_{t,ih} E_{t,jh} \right\|^2 \\
&= O_P\left(T^{-2} p^{-2\delta_r, k_r} q^{-2\delta_{c,1}} (Tq + T^2 q^2 p^{-\delta_r, k_r})\right) = O_P\left(T^{-1} p^{-2\delta_r, k_r} q^{1-2\delta_{c,1}} + p^{-3\delta_r, k_r} q^{2-2\delta_{c,1}}\right).
\end{aligned} \tag{S.24}$$

Consider now $\mathcal{I}_4$ and $\mathcal{I}_5$. From the proof of (S.3) in Lemma 2,

$$\begin{aligned}
\|\mathcal{I}_4\|^2 &\leq \frac{1}{T^2 p^2 q^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p)_{ij} \right\|^2 \\
&\leq \frac{1}{T^2 p^2 q^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left(\sum_{i=1}^p \|\mathbf{Q}_{r,i}\|^2 \right) \sum_{i=1}^p \left(\sum_{t=1}^T \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p \right)_{ij}^2 \\
&= O_P\left(T^{-2} p^{-2-2\delta_r, k_r} q^{-2-2\delta_{c,1}}\right) \cdot \left\| \left(\sum_{t=1}^T \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p \right)_{\cdot j} \right\|^2 \\
&= O_P\left(T^{-2} p^{-2-2\delta_r, k_r} q^{-2-2\delta_{c,1}}\right) \cdot \sum_{k=1}^p \mathbb{E}\left\{ \left[\sum_{t=1}^T \left(\sum_{i=1}^p \sum_{u=1}^q E_{t,iu} \right) \sum_{h=1}^q E_{t,kh} \right]^2 \right\} \\
&= O_P\left(T^{-2} p^{-2-2\delta_r, k_r} q^{-2-2\delta_{c,1}} (Tp^2 q^2 + T^2 pq^2)\right) \\
&= O_P\left(T^{-1} p^{-2\delta_r, k_r} q^{-2\delta_{c,1}} + p^{-1-2\delta_r, k_r} q^{-2\delta_{c,1}}\right),
\end{aligned}$$

where we used Lemma 3 in the third line and (S.16) in the second last equality. Similarly, we have

$$\begin{aligned}
\|\mathcal{I}_5\|^2 &\leq \frac{1}{T^2 p^2 q^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_q \mathbf{E}'_t)_{ij} \right\|^2 \\
&= O_P\left(T^{-2} p^{-2-2\delta_r, k_r} q^{-2-2\delta_{c,1}}\right) \cdot \left\| \left(\sum_{t=1}^T \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p \right)_{\cdot j} \right\|^2 \\
&= O_P\left(T^{-2} p^{-2-2\delta_r, k_r} q^{-2-2\delta_{c,1}}\right) \cdot p \cdot \mathbb{E}\left\{ \left[\sum_{t=1}^T \left(\sum_{i=1}^p \sum_{u=1}^q E_{t,iu} \right) \sum_{h=1}^q E_{t,jh} \right]^2 \right\} \\
&= O_P\left(T^{-2} p^{-2-2\delta_r, k_r} q^{-2-2\delta_{c,1}} (Tp^2 q^2 + T^2 pq^2)\right) \\
&= O_P\left(T^{-1} p^{-2\delta_r, k_r} q^{-2\delta_{c,1}} + p^{-1-2\delta_r, k_r} q^{-2\delta_{c,1}}\right),
\end{aligned}$$

where we used again Lemma 3 in the third line and (S.16) in the second last equality.

Consider now $\mathcal{I}_6$, note first from Assumptions (E1) and (E2),

$$E_{t,hu} = \sum_{w \geq 0} a_{e,w} \mathbf{A}'_{e,r,h} \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,u} + (\boldsymbol{\Sigma}_\epsilon)_{hu} \sum_{w \geq 0} a_{e,w} (\mathbf{X}_{e,t-w})_{hu}.$$

Thus, we have

$$\begin{aligned} \text{cov}(E_{t,hu}, E_{s,vl}) &= \mathbb{E}[\mathbf{A}'_{e,r,h} \mathbf{F}_{e,t} \mathbf{A}_{e,c,u} \mathbf{A}'_{e,c,l} \mathbf{F}'_{e,s} \mathbf{A}_{e,r,v}] + \mathbb{1}_{\{h=v\}} \mathbb{1}_{\{u=l\}} \cdot (\boldsymbol{\Sigma}_\epsilon)_{hu}^2 \cdot \mathbb{E}[\boldsymbol{\epsilon}'_{t,u} \boldsymbol{\epsilon}_{s,l}] \\ &= \mathbf{A}'_{e,c,l} \mathbf{A}_{e,c,u} \mathbf{A}'_{e,r,h} \mathbf{A}_{e,r,v} \cdot \sum_{w \geq 0} a_{e,w} a_{e,w+|t-s|} + \mathbb{1}_{\{h=v\}} \mathbb{1}_{\{u=l\}} \cdot (\boldsymbol{\Sigma}_\epsilon)_{hu}^2 \sum_{w \geq 0} a_{e,w} a_{e,w+|t-s|}. \end{aligned}$$

Hence if we fix $t \in [T]$, $h \in [p]$, $u \in [q]$, then for any deterministic vectors $\mathbf{n} \in \mathbb{R}^{k_r}$ and $\mathbf{g}, \mathbf{j} \in \mathbb{R}^{k_c}$, we have

$$\begin{aligned} \sum_{s=1}^T \sum_{v=1}^p \sum_{l=1}^q \mathbb{E}[E_{t,hu} \mathbf{n}' \mathbf{F}_t \mathbf{g} \cdot E_{s,vl} \mathbf{j}' \mathbf{F}'_s \mathbf{n}] &= \sum_{s=1}^T \sum_{v=1}^p \sum_{l=1}^q \text{cov}(E_{t,hu}, E_{s,vl}) \cdot \mathbb{E}[\mathbf{n}' \mathbf{F}_t \mathbf{g} \mathbf{j}' \mathbf{F}'_s \mathbf{n}] \\ &= \sum_{v=1}^p \sum_{l=1}^q \left[O(\mathbf{A}'_{e,c,l} \mathbf{A}_{e,c,u} \mathbf{A}'_{e,r,h} \mathbf{A}_{e,r,v}) \cdot \sum_{w \geq 0} \sum_{m \geq 0} \sum_{s=1}^T a_{e,w} a_{e,w+|t-s|} a_{f,m} a_{f,m+|t-s|} \right. \\ &\quad \left. + O(\mathbb{1}_{\{h=v\}} \mathbb{1}_{\{u=l\}} \cdot (\boldsymbol{\Sigma}_\epsilon)_{hu}^2) \cdot \sum_{w \geq 0} \sum_{m \geq 0} \sum_{s=1}^T a_{e,w} a_{e,w+|t-s|} a_{f,m} a_{f,m+|t-s|} \right] \\ &= \sum_{v=1}^p \sum_{l=1}^q O(\mathbf{A}'_{e,c,l} \mathbf{A}_{e,c,u} \mathbf{A}'_{e,r,h} \mathbf{A}_{e,r,v} + \mathbb{1}_{\{h=v\}} \mathbb{1}_{\{u=l\}} \cdot (\boldsymbol{\Sigma}_\epsilon)_{hu}^2) = O(1), \end{aligned}$$

where for the second last equality, we argue for the first term in the second last line only, as the second term could be shown similarly:

$$\begin{aligned} \sum_{w \geq 0} \sum_{m \geq 0} \sum_{s=1}^T a_{e,w} a_{e,w+|t-s|} a_{f,m} a_{f,m+|t-s|} &= \sum_{w \geq 0} \sum_{m \geq 0} a_{e,w} a_{f,m} \sum_{s=1}^T a_{e,w+|t-s|} a_{f,m+|t-s|} \\ &\leq \sum_{w \geq 0} \sum_{m \geq 0} |a_{e,w}| \cdot |a_{f,m}| \cdot \left(\sum_{s=1}^T a_{e,w+|t-s|}^2 \right)^{1/2} \left(\sum_{s=1}^T a_{f,m+|t-s|}^2 \right)^{1/2} \leq \sum_{w \geq 0} \sum_{m \geq 0} |a_{e,w}| \cdot |a_{f,m}| \leq c^2, \end{aligned}$$

where the constant c is from Assumptions (F1) and (E2). Finally,

$$\begin{aligned} \mathbb{E}\left\{ \left(\sum_{h=1}^p \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}'_t)_{ih} \right)^2 \right\} &= \mathbb{E}\left\{ \left(\sum_{u=1}^q \sum_{h=1}^p \sum_{t=1}^T E_{t,hu} \mathbf{A}'_{e,c,u} \mathbf{F}'_t \mathbf{A}_{r,i} \right)^2 \right\} \\ &= \sum_{t=1}^T \sum_{h=1}^p \sum_{u=1}^q \sum_{s=1}^T \sum_{v=1}^p \sum_{l=1}^q \mathbb{E}\left[E_{t,hu} \mathbf{A}'_{e,c,u} \mathbf{F}_t \mathbf{A}_{c,u} \cdot E_{s,vl} \mathbf{A}'_{e,c,l} \mathbf{F}'_s \mathbf{A}_{r,i} \right] = O(Tpq). \end{aligned} \tag{S.25}$$

Thus, we have

$$\begin{aligned} \|\mathcal{I}_6\|^2 &\leq \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_p)_{ij} \right\|^2 \\ &= \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{h=1}^p \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}'_t)_{ih} (\mathbf{1}_p \mathbf{1}'_p)_{hj} \right\|^2 \\ &\leq \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \|\mathbf{Q}_r\|_F^2 \cdot \sum_{i=1}^p \left(\sum_{h=1}^p \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}'_t)_{ih} \right)^2 \\ &= O_P(T^{-2} p^{-2-2\delta_{r,k_r}} q^{-2\delta_{c,1}} \cdot T p^2 q) = O_P(T^{-1} p^{-2\delta_{r,k_r}} q^{1-2\delta_{c,1}}), \end{aligned}$$

where we used Cauchy-Schwarz inequality in the third line, both Lemma 3 and (S.25) in the second last equality. Similarly, we have the following for $\mathcal{I}_7$,

$$\begin{aligned}
\|\mathcal{I}_7\|^2 &\leq \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{C}_t')_{ij} \right\|^2 \\
&= \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{h=1}^p \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}_p \mathbf{1}_p')_{ih} (\mathbf{E}_t \mathbf{C}_t')_{hj} \right\|^2 \\
&\leq \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \|\mathbf{Q}_r\|_F^2 \cdot p \left(\sum_{h=1}^p \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}_t')_{jh} \right)^2 \\
&= O_P(T^{-2} p^{-2-2\delta_{r,k_r}} q^{-2\delta_{c,1}} \cdot T p^2 q) = O_P(T^{-1} p^{-2\delta_{r,k_r}} q^{1-2\delta_{c,1}}).
\end{aligned}$$

For $\mathcal{I}_8$ and $\mathcal{I}_9$, their rates can be shown to be the same as that for $\mathcal{I}_3$ by the following,

$$\begin{aligned}
\|\mathcal{I}_8\|^2 &\leq \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p')_{ij} \right\|^2 \\
&= \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{h=1}^p \sum_{i=1}^p \sum_{l=1}^q \sum_{t=1}^T \mathbf{Q}_{r,i} \cdot E_{t,il} E_{t,hl} \right\|^2 \\
&\leq \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot p \sum_{h=1}^p \left\| \sum_{i=1}^p \sum_{l=1}^q \sum_{t=1}^T \mathbf{Q}_{r,i} \cdot E_{t,il} E_{t,hl} \right\|^2 \\
&= O_P(T^{-2} p^{-2-2\delta_{r,k_r}} q^{-2\delta_{c,1}} \cdot p^2 (Tq + T^2 q^2 p^{-\delta_{r,k_r}})) = O_P(T^{-1} p^{-2\delta_{r,k_r}} q^{1-2\delta_{c,1}} + p^{-3\delta_{r,k_r}} q^{2-2\delta_{c,1}}),
\end{aligned}$$

where we used Lemma 3 and (S.24) in the second last equality. The proof for $\|\mathcal{I}_9\|^2$ is similar to the above by using the proof of (S.24) previously and omitted here.

For $\mathcal{I}_{10}$, first observe from the proof of (S.16), we also have for any $j \in [p]$,

$$\sum_{i=1}^p \mathbb{E} \left\{ \left(\sum_{t=1}^T \sum_{l=1}^q E_{t,il} \sum_{h=1}^q E_{t,jh} \right)^2 \right\} = O(Tpq^2 + T^2 q^2),$$

then together with Lemma 3, it holds that

$$\begin{aligned}
\|\mathcal{I}_{10}\|^2 &\leq \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t')_{ij} \right\|^2 \\
&= \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T \sum_{l=1}^q E_{t,il} \sum_{h=1}^q E_{t,jh} \right\|^2 \\
&\leq \frac{1}{T^2 p^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \|\mathbf{Q}_r\|_F^2 \cdot \sum_{i=1}^p \left(\sum_{t=1}^T \sum_{l=1}^q E_{t,il} \sum_{h=1}^q E_{t,jh} \right)^2 \\
&= O_P(T^{-2} p^{-2-2\delta_{r,k_r}} q^{-2\delta_{c,1}} \cdot (Tpq^2 + T^2 q^2)) = O_P(T^{-1} p^{-1-2\delta_{r,k_r}} q^{2-2\delta_{c,1}} + p^{-2-2\delta_{r,k_r}} q^{2-2\delta_{c,1}}).
\end{aligned}$$

Consider now $\mathcal{I}_{11}$, we have

$$\begin{aligned}
\|\mathcal{I}_{11}\|^2 &\leq \frac{1}{T^2 p^4 q^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \cdot \sum_{t=1}^T (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p)^2 \right\|^2 \\
&\leq \frac{1}{T^2 p^4 q^2} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \|\mathbf{Q}_r\|_F^2 \cdot p \left[\sum_{t=1}^T (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p)^2 \right]^2 \\
&= O_P(T^{-2} p^{-4-2\delta_{r,k_r}} q^{-2-2\delta_{c,1}} \cdot T^2 p^3 q^2) = O_P(p^{-1-2\delta_{r,k_r}} q^{-2\delta_{c,1}}),
\end{aligned}$$

where we used Lemma 3 and the rate from (S.2) in the second last equality.

For $\mathcal{I}_{12}$, we have

$$\begin{aligned}\|\mathcal{I}_{12}\|^2 &\leq \frac{1}{T^2 p^4} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \left\| \sum_{i=1}^p \mathbf{Q}_{r,i} \sum_{t=1}^T \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \right\|^2 \\ &\leq \frac{1}{T^2 p^4} \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\mathbf{H}_r\|_F^2 \cdot \|\mathbf{Q}_r\|_F^2 \cdot p \left(\sum_{t=1}^T \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \right)^2 \\ &= O_P(T^{-2} p^{-4-2\delta_{r,kr}} q^{-2\delta_{c,1}} \cdot T^2 p^3 q^2) = O_P(p^{-1-2\delta_{r,kr}} q^{2-2\delta_{c,1}}),\end{aligned}$$

where the last line used the following result which can be shown similar to (S.9),

$$\begin{aligned}\mathbb{E}\left\{\left(\sum_{t=1}^T \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p\right)^2\right\} &= \mathbb{E}\left\{\left(\sum_{t=1}^T \sum_{h=1}^q \sum_{i=1}^p \sum_{j=1}^p E_{t,ih} E_{t,jh}\right)^2\right\} \\ &= \sum_{t=1}^T \sum_{h=1}^q \sum_{i=1}^p \sum_{j=1}^p \sum_{s=1}^T \sum_{l=1}^q \sum_{m=1}^p \sum_{n=1}^p \text{cov}(E_{t,ih} E_{t,jh}, E_{s,ml} E_{s,nl}) + \left(\sum_{t=1}^T \sum_{h=1}^q \sum_{i=1}^p \sum_{j=1}^p \mathbb{E}[E_{t,ih} E_{t,jh}]\right)^2 \\ &= O(T p^2 q^2 + T^2 p^2 q^2) = O(T^2 p^2 q^2).\end{aligned}$$

Lastly, $\|\mathcal{I}_{13}\|^2$ is dominated by the terms from $\mathcal{I}_1$ to $\mathcal{I}_{12}$ using Cauchy-Schwarz inequality and Theorem 2. We require the term $\mathcal{I}_1$ to be truly dominating by using Assumption (AD1) and we equivalently compare the rates without the term $\widehat{\mathbf{D}}_r^{-1}$. Notice the rates for $\|\mathcal{I}_2\|^2$, $\|\mathcal{I}_4\|^2$, $\|\mathcal{I}_5\|^2$, $\|\mathcal{I}_6\|^2$, $\|\mathcal{I}_7\|^2$, $\|\mathcal{I}_{10}\|^2$, $\|\mathcal{I}_{11}\|^2$ and $\|\mathcal{I}_{12}\|^2$ are bounded above by the rate for $\|\mathcal{I}_3\|^2$ which is the same as $\|\mathcal{I}_8\|^2$ and $\|\mathcal{I}_9\|^2$. Thus, it suffices to consider the following ratio as $p, q, T \rightarrow \infty$,

$$\|\mathcal{I}_3\|^2 / \|\mathcal{I}_1\|^2 = O_P(1/p^{\delta_{r,1}} + Tq/p^{\delta_{r,1}+\delta_{r,kr}}) = o_P(1),$$

by the rate assumption $Tq = o(p^{\delta_{r,1}+\delta_{r,kr}})$. Therefore, $\mathcal{I}_1$ is dominating over other terms in (S.21) and hence we have

$$\begin{aligned}\widehat{\mathbf{Q}}_{r,j\cdot} - \mathbf{H}_r \mathbf{Q}_{r,j\cdot} &= \mathcal{I}_1 + o_P(1) = \frac{1}{T} \widehat{\mathbf{D}}_r^{-1} \sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i\cdot} \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}_t')_{ij} + o_P(1) \\ &\xrightarrow{P} \frac{1}{T} \mathbf{D}_r^{-1} \mathbf{H}_r^* \sum_{i=1}^p \mathbf{Q}_{r,i\cdot} \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}_t')_{ij},\end{aligned}\tag{S.26}$$

where the last line used Lemma 4 and Lemma 5 in which $\mathbf{D}_r$ and $\mathbf{H}_r^*$ are defined, respectively. In the rest of the proof, we show

$$\sqrt{T\omega_B} \cdot \frac{1}{T} \mathbf{D}_r^{-1} \mathbf{H}_r^* \sum_{i=1}^p \mathbf{Q}_{r,i\cdot} \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}_t')_{ij} \xrightarrow{D} \mathcal{N}(\mathbf{0}, T^{-1} \omega_B \cdot \mathbf{D}_r^{-1} \mathbf{H}_r^* \mathbf{\Xi}_{r,j} (\mathbf{H}_r^*)' \mathbf{D}_r^{-1}),$$

with $\omega_B := p^{2\delta_{r,kr}-\delta_{r,1}} q^{2\delta_{c,1}-1}$ and $\mathbf{\Xi}_{r,j} := \text{plim}_{p,q,T \rightarrow \infty} \text{Var}(\sum_{i=1}^p \mathbf{Q}_{r,i\cdot} \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}_t')_{ij})$. We will adapt the central limit theorem for α -mixing processes as depicted in Theorem 2.21 in Fan and Yao (2003). First, define $\mathbf{B}_{j,t} := \sqrt{\omega_B} \cdot \mathbf{D}_r^{-1} \mathbf{H}_r^* \sum_{i=1}^p \mathbf{Q}_{r,i\cdot} (\mathbf{C}_t \mathbf{E}_t')_{ij}$, and also let $\mathbf{b}_{e,t} := \sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w}$, $\mathbf{b}_{\epsilon,il,t} := \sum_{w \geq 0} a_{\epsilon,w} (\mathbf{X}_{\epsilon,t-w})_{il}$ and $\mathbf{b}_{f,t} := \sum_{w \geq 0} a_{f,w} \mathbf{X}_{f,t-w}$ which are independent of each other by Assumption (E2).

Since we may write $\mathbf{B}_{j,t} = h(\mathbf{b}_{e,t}, (\mathbf{b}_{\epsilon,il,t})_{i \in [p], l \in [q]}, \mathbf{b}_{f,t})$ for some function h , we conclude $\mathbf{B}_{j,t}$ is α -mixing using Theorem 5.2 in Bradley (2005). Observe that $\mathbb{E}[\mathbf{B}_{j,t}] = \mathbf{0}$, and we show in the following that

there exists an $m > 2$ such that $\mathbb{E}[\|\mathbf{B}_{j,t}\|^m] \leq C$ for some constant C ,

$$\begin{aligned} \mathbb{E}[\|\mathbf{B}_{j,t}\|^m] &\leq \omega_B^{m/2} \cdot \|\mathbf{D}_r^{-1}\|_F^m \cdot \|\mathbf{H}_r^*\|_F^m \cdot \|\mathbf{Q}_r\|_F^m \cdot \mathbb{E}\left\{\left[\sum_{i=1}^p \left(\sum_{l=1}^q E_{t,jl} \mathbf{A}'_{c,h} \mathbf{F}'_t \mathbf{A}_{r,i}\right)^2\right]^{m/2}\right\} \\ &= O\left((p^{2\delta_{r,k_r}} q^{2\delta_{c,1}})^{m/2}\right) \cdot \|\mathbf{D}_r^{-1}\|_F^m = O(1), \end{aligned}$$

where we used Lemma 1.2 and the definition of ω_B in the second last equality, and Lemma 4 in the last equality. Theorem 2.21 in Fan and Yao (2003) then applies, and hence

$$\sqrt{T\omega_B} \cdot \frac{1}{T} \mathbf{D}_r^{-1} \mathbf{H}_r^* \sum_{i=1}^p \mathbf{Q}_{r,i} \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}'_t)_{ij} = \frac{1}{\sqrt{T}} \sum_{t=1}^T \mathbf{B}_{j,t} \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, T^{-1} \omega_B \cdot \mathbf{D}_r^{-1} \mathbf{H}_r^* \mathbf{\Xi}_{r,j} (\mathbf{H}_r^*)' \mathbf{D}_r^{-1}).$$

Together with (S.26), we arrive at

$$(Tp^{2\delta_{r,k_r}-\delta_{r,1}} q^{2\delta_{c,1}-1})^{1/2} \cdot (\hat{\mathbf{Q}}_{r,j} - \mathbf{H}_r \mathbf{Q}_{r,j}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, T^{-1} p^{2\delta_{r,k_r}-\delta_{r,1}} q^{2\delta_{c,1}-1} \cdot \mathbf{D}_r^{-1} \mathbf{H}_r^* \mathbf{\Xi}_{r,j} (\mathbf{H}_r^*)' \mathbf{D}_r^{-1}). \quad (\text{S.27})$$

This completes the proof of Theorem 5. $\square$

Proof of Theorem 6. We only prove the scenario for $Tq = o(p^{\delta_{r,1}+\delta_{r,k_r}})$, which is on showing the constructed estimator for the row loading estimator is consistent. Note $\hat{\mathbf{D}}_r$ consistently estimates $\mathbf{D}_r$ by Lemma 4, and $\mathbf{H}_r$ consistently estimates $\mathbf{H}_r^*$ by Lemma 5. Then observe that

$$\begin{aligned} \mathbf{H}_r \text{Var}\left\{\sum_{i=1}^p \mathbf{Q}_{r,i} \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}'_t)_{ij}\right\} \mathbf{H}'_r &= \text{Var}\left\{\sum_{i=1}^p \mathbf{H}_r \mathbf{Q}_{r,i} \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}'_t)_{ij}\right\} \\ &= \text{Var}\left\{\sum_{i=1}^p \left(T^{-1} \hat{\mathbf{D}}_r^{-1} \hat{\mathbf{Q}}'_r \sum_{s=1}^T \mathbf{F}_{Z,s} \mathbf{Q}'_c \mathbf{Q}_c \mathbf{F}'_{Z,s}\right) \mathbf{Q}_{r,i} \sum_{t=1}^T (\mathbf{C}_t \mathbf{E}'_t)_{ij}\right\} \\ &= \text{Var}\left\{\sum_{t=1}^T \sum_{i=1}^p \left(T^{-1} \hat{\mathbf{D}}_r^{-1} \hat{\mathbf{Q}}'_r \sum_{s=1}^T \mathbf{C}_s \mathbf{C}_{s,i}\right) (\mathbf{C}_t \mathbf{E}'_t)_{ij}\right\}. \end{aligned}$$

By Theorem 2, $(\hat{\mu}_t, \hat{\alpha}_t, \hat{\beta}_t)$ is consistent for $(\mu_t, \alpha_t, \beta_t)$. By Theorem 3 and the rate assumption in the statement of Theorem 6, $\hat{\mathbf{C}}_t$ is consistent for $\mathbf{C}_t$ and hence $\hat{\mathbf{E}}_t$ is consistent for $\mathbf{E}_t$. Thus, we conclude that $\hat{\Sigma}_{r,j}^{HAC}$ is estimating $\mathbf{H}_r \mathbf{\Xi}_{r,j} \mathbf{H}'_r$ consistently (Newey and West (1987)) and hence result 1 is implied. Result 2 then follows, and results 3 and 4 can be shown similarly (details omitted). This completes the proof of the Theorem 6. $\square$

Proof of Theorem 7. Combining (3.4) and (3.6), we have

$$\begin{aligned} \hat{\mathbf{E}}_t &= \hat{\mathbf{L}}_t - \hat{\mathbf{C}}_t = \mathbf{M}_p \mathbf{Y}_t \mathbf{M}_q - \hat{\mathbf{C}}_t = \mathbf{M}_p \mathbf{C}_t \mathbf{M}_q + \mathbf{M}_p \mathbf{E}_t \mathbf{M}_q - \hat{\mathbf{C}}_t \\ &= (\mathbf{C}_t - \hat{\mathbf{C}}_t) + \mathbf{M}_p \mathbf{A}_{e,r} \mathbf{F}_{e,t} \mathbf{A}'_{e,c} \mathbf{M}_q + \mathbf{M}_p (\Sigma_\epsilon \circ \epsilon_t) \mathbf{M}_q, \end{aligned}$$

where the second line comes from (IC1) being satisfied, so that $\mathbf{M}_p \mathbf{A}_r = \mathbf{A}_r$ and $\mathbf{M}_q \mathbf{A}_c = \mathbf{A}_c$. Hence

$$\begin{aligned}
q^{-1}(\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t')_{ii} - q^{-1} \sum_{j=1}^q \Sigma_{\epsilon, ij}^2 &= \sum_{i=1}^6 I_i, \text{ where} \\
I_1 &:= q^{-1} \{ \mathbf{M}_p (\Sigma_\epsilon \circ \epsilon_t) \mathbf{M}_q (\Sigma_\epsilon \circ \epsilon_t)' \mathbf{M}_p \}_{ii} - q^{-1} \sum_{j=1}^q \Sigma_{\epsilon, ij}^2, \\
I_2 &:= q^{-1} \{ (\widehat{\mathbf{C}}_t - \mathbf{C}_t) (\widehat{\mathbf{C}}_t - \mathbf{C}_t)' \}_{ii}, \\
I_3 &:= q^{-1} \{ \mathbf{M}_p \mathbf{A}_{e,r} \mathbf{F}_{e,t} \mathbf{A}_{e,c}' \mathbf{M}_q \mathbf{A}_{e,c} \mathbf{F}_{e,t}' \mathbf{A}_{e,r}' \mathbf{M}_p \}_{ii}, \\
I_4 &:= O_P(q^{-1} \{ (\mathbf{C}_t - \widehat{\mathbf{C}}_t) \mathbf{M}_q \mathbf{A}_{e,c} \mathbf{F}_{e,t}' \mathbf{A}_{e,r}' \mathbf{M}_p \}_{ii}), \\
I_5 &:= O_P(q^{-1} \{ (\mathbf{C}_t - \widehat{\mathbf{C}}_t) \mathbf{M}_q (\Sigma_\epsilon \circ \epsilon_t)' \mathbf{M}_p \}_{ii}), \\
I_6 &:= O_P(q^{-1} \{ \mathbf{M}_p \mathbf{A}_{e,r} \mathbf{F}_{e,t} \mathbf{A}_{e,c}' \mathbf{M}_q (\Sigma_\epsilon \circ \epsilon_t)' \mathbf{M}_p \}_{ii}).
\end{aligned} \tag{S.28}$$

By an assumption in the statement of the theorem, we have $I_2 = o_P(q^{-1})$. Since $\|\mathbf{M}_p \mathbf{A}_{e,r}\|_1 \leq \|\mathbf{M}_p\|_1 \|\mathbf{A}_{e,r}\|_1 = O(1)$, with the finiteness of k_r and k_c , we have $I_3, I_6 = o_P(q^{-1})$, and by the Cauchy-Schwarz inequality, $I_4 = O_P(I_2^{1/2} \cdot I_6^{1/2}) = o_P(q^{-1})$. Writing $\eta_{t,ij} := (\Sigma_\epsilon \circ \epsilon_t)_{ij}$, we can expand

$$\begin{aligned}
I_1 &= \frac{1}{q} \sum_{j=1}^q \Sigma_{\epsilon, ij}^2 (\epsilon_{t,ij}^2 - 1) - \frac{2}{q} \sum_{j=1}^q \left(\frac{1}{p} \sum_{\ell=1}^p \eta_{t,\ell j} \right) \eta_{t,ij} \\
&\quad + \frac{2}{q} \left(\frac{\mathbf{1}_p' \boldsymbol{\eta}_t \mathbf{1}_q}{pq} \right) \sum_{j=1}^q \eta_{t,ij} + \frac{1}{q} \sum_{j=1}^q \left(\frac{1}{p} \sum_{i=1}^p \eta_{t,ij} \right)^2 - \left(\frac{\mathbf{1}_p' \boldsymbol{\eta}_t \mathbf{1}_q}{pq} \right)^2 - \left(\frac{1}{q} \sum_{j=1}^q \eta_{t,ij} \right)^2, \\
&= \frac{1}{q} \sum_{j=1}^q \Sigma_{\epsilon, ij}^2 (\epsilon_{t,ij}^2 - 1) - \frac{2}{pq} \sum_{j=1}^q \eta_{t,ij}^2 - \frac{2}{pq} \sum_{j=1}^q \sum_{\ell \neq i} \eta_{t,\ell j} \eta_{t,ij} \\
&\quad + O_P(q^{-1} p^{-1/2}) + O_P(p^{-1} q^{-1/2}) + O_P(p^{-1} q^{-1}) + O_P(q^{-1}) \\
&= \frac{1}{q} \sum_{j=1}^q \Sigma_{\epsilon, ij}^2 (\epsilon_{t,ij}^2 - 1) + O_P(p^{-1} q^{-1/2}) + O_P(p^{-1/2} q^{-1/2}) \\
&\quad + O_P(q^{-1} p^{-1/2}) + O_P(p^{-1} q^{-1/2}) + O_P(p^{-1} q^{-1}) + O_P(q^{-1}) \\
&= \frac{1}{q} \sum_{j=1}^q \Sigma_{\epsilon, ij}^2 (\epsilon_{t,ij}^2 - 1) (1 + o_P(1)),
\end{aligned} \tag{S.29}$$

where all rates of convergence above are obtained from applying the Markov inequality. Hence

$$\begin{aligned}
\frac{(\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t')_{ii} - \sum_{j=1}^q \Sigma_{\epsilon, ij}^2}{\sqrt{\sum_{j=1}^q \Sigma_{\epsilon, ij}^4 \text{Var}(\epsilon_{t,ij}^2)}} &= \frac{q^{-1}(\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t')_{ii} - q^{-1} \sum_{j=1}^q \Sigma_{\epsilon, ij}^2}{\sqrt{q^{-2} \sum_{j=1}^q \text{Var}(\epsilon_{t,ij}^2) \Sigma_{\epsilon, ij}^4}} \\
&= \frac{q^{-1} \sum_{j=1}^q \Sigma_{\epsilon, ij}^2 (\epsilon_{t,ij}^2 - 1) (1 + o_P(1))}{\sqrt{q^{-2} \sum_{j=1}^q \text{Var}(\epsilon_{t,ij}^2) \Sigma_{\epsilon, ij}^4}} \\
&\xrightarrow{\mathcal{D}} Z_{i,t} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1),
\end{aligned}$$

where the last line follows from Theorem 1 in Ayvazyan and Ulyanov (2023), with

$$Z_{i,t} := \frac{q^{-1} \sum_{j=1}^q \Sigma_{\epsilon, ij}^2 (\epsilon_{t,ij}^2 - 1)}{\sqrt{q^{-2} \sum_{j=1}^q \text{Var}(\epsilon_{t,ij}^2) \Sigma_{\epsilon, ij}^4}}, \quad i \in [p],$$

so that we can easily see that the $Z_{i,t}$'s are independent of each other by Assumption (E1). The proof for

$(\widehat{\mathbf{E}}_t \widehat{\mathbf{E}}_t')_{ii}$ completes since from the calculations for I_1 , we see that

$$I_5 = O_P(I_2^{1/2} \cdot 1) = o_P(q^{-1/2}).$$

For $(\check{\mathbf{E}}_t \check{\mathbf{E}}'_t)_{ii}$ under H_0 , note that we have

$$\check{\mathbf{E}}_t = (\mathbf{C}_t - \check{\mathbf{C}}_t) + \mathbf{A}_{e,r} \mathbf{F}_{e,t} \mathbf{A}'_{e,c} + (\boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\epsilon}_t),$$

with the rate for $\check{C}_{t,ij} - C_{t,ij}$ the same as that for $\hat{C}_{t,ij} - C_{t,ij}$ since the estimation procedure for FM is essentially the same with the same assumptions on the factor loadings (apart from (IC1) which is not important for FM), the factors and the noise. Hence the proof we employed so far can be replicated with $\mathbf{M}_p$ and $\mathbf{M}_q$ replaced by the corresponding sized identity matrices, and we arrive at the same conclusion with the same $Z_{i,t}$'s.

For $(\hat{\mathbf{E}}'_t \hat{\mathbf{E}}_t)_{jj}$ under both H_0 and H_1 and $(\check{\mathbf{E}}'_t \check{\mathbf{E}}_t)_{jj}$ under H_0 , the proofs are parallel to that for $(\hat{\mathbf{E}}_t \hat{\mathbf{E}}'_t)_{ii}$, and we omit them here. $\square$

Proof of Theorem 8.

We provide the details of proof for the inequality concerning $\mathbb{P}_{y,\alpha}(y_{\alpha,t} \geq \hat{q}_{y,\alpha}(\theta))$. The other one involves similar details and its proof is omitted.

Firstly, using triangle inequality, consider

$$\begin{aligned} \max_{t \in [T]} |x_{\alpha,t} - y_{\alpha,t}| &\leq \max_{i \in [p], t \in [T]} |q^{-1}(\hat{E}_t \hat{E}'_t)_{ii} - q^{-1}(\check{E}_t \check{E}'_t)_{ii}| \\ &\leq \max_{i \in [p], t \in [T]} |I_1^*| + \sum_{\ell=2}^6 \max_{i \in [p], t \in [T]} (|I_\ell| + |\tilde{I}_\ell|), \end{aligned}$$

where I_ℓ , $\ell = 2, \dots, 6$ is defined as in (S.28), and $\tilde{I}_\ell$ for $\ell = 2, \dots, 6$ is defined exactly the same as I_ℓ , except that $\mathbf{M}_p$ and $\mathbf{M}_q$ in the definitions of the I_ℓ 's are replaced by identity matrices of appropriate sizes, and $\hat{\mathbf{C}}_t$ is replaced by $\check{\mathbf{C}}_t$. The definition of I_1^* is the same as I_1 in (S.28), except that the term $q^{-1} \sum_{j=1}^q \Sigma_{\epsilon,ij}^2 (\epsilon_{t,ij}^2 - 1)$ is absent in (S.29).

With pervasive factors, the uniform error rates for $\hat{\mathbf{C}}_t$ (and $\check{\mathbf{C}}_t$, since the form of $\check{\mathbf{C}}_t$ is the same as that in (3.5), with rates for $\check{\mathbf{Q}}_r$ and $\check{\mathbf{Q}}_c$ similar to $\hat{\mathbf{Q}}_r$ and $\hat{\mathbf{Q}}_c$ respectively) from Lemma 9 is

$$\begin{aligned} &\max_{i \in [p], j \in [q], t \in [T]} \{|\hat{C}_{t,ij} - \check{C}_{t,ij}|, |\check{C}_{t,ij} - \check{C}_{t,ij}|\} \\ &= O_P\{(pq)^{-1/2} + (Tq)^{-1/2} + (Tp)^{-1/2} + p^{-1} + q^{-1} \log(T) \log(p) \log(q)\}. \end{aligned} \quad (\text{S.30})$$

From the proof of Theorem 7, we can see that I_1^* has faster convergence rate than other terms, and in fact I_5 (and hence $\tilde{I}_5$) has the slowest rate of convergence for a fixed indices $t \in [T]$ and $i \in [p]$. Taking maximum over all possible indices, using Assumption (E3) and (S.30), we thus have

$$\begin{aligned} \max_{t \in [T]} |x_{\alpha,t} - y_{\alpha,t}| &= O_P(I_5) = O_P\left(\max_{t \in [T], i \in [p], j \in [q]} |C_{t,ij} - \hat{C}_{t,ij}| \cdot \max_{t \in [T], i \in [p], j \in [q]} |(\boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\epsilon}_t)_{ij}|\right) \\ &= O_P\{((pq)^{-1/2} + (Tq)^{-1/2} + (Tp)^{-1/2} + p^{-1} + q^{-1}) \log^2(T) \log^2(p) \log^2(q)\}. \end{aligned} \quad (\text{S.31})$$

Next, we assess the approximate ‘‘gap’’ size of the $x_{t,\alpha}$'s over $t \in [T]$. To this end, using Theorem 7, and the fact that $\sum_{j=1}^q \text{Var}(\epsilon_{t,ij}^2) \Sigma_{\epsilon,ij}^4$ has order q uniformly over $i \in [p]$, we have

$$x_{\alpha,t} \asymp_P \max_{i \in [p]} \left\{ q^{-1} \sum_{j=1}^q \Sigma_{\epsilon,ij}^2 + \frac{1}{\sqrt{q}} \max_{i \in [p]} Z_{i,t} \right\} \asymp_P \max_{i \in [p]} \left\{ q^{-1} \sum_{j=1}^q \Sigma_{\epsilon,ij}^2 \right\} + \frac{\log(p)}{\sqrt{q}}, \quad (\text{S.32})$$

showing that the ‘‘gap’’ between two ordered $x_{\alpha,t}$'s is $O_P(\log(p)/(Tq^{1/2}))$.

With the “gap” size and uniform error in (S.31), consider

$$\begin{aligned}
\sup_{c \in \mathbb{R}} |\mathbb{F}_{y,\alpha}(c) - \mathbb{F}_{x,\alpha}(c)| &= \sup_{c \in \mathbb{R}} \left| \frac{1}{T} \sum_{t=1}^T [\mathbf{1}\{y_{\alpha,t} \leq c\} - \mathbf{1}\{x_{\alpha,t} \leq c\}] \right| \\
&\leq \sup_{c \in \mathbb{R}} \left| \frac{1}{T} \sum_{t=1}^T [\mathbf{1}\{x_{\alpha,t} \pm \max_{t \in [T]} |x_{\alpha,t} - y_{\alpha,t}| \leq c\} - \mathbf{1}\{x_{\alpha,t} \leq c\}] \right| \\
&= O_P \left(\frac{1}{T} \max_{t \in [T]} |x_{\alpha,t} - y_{\alpha,t}| / \left(\frac{\log(p)}{T\sqrt{q}} \right) \right) \\
&= O_P \left\{ \left(\frac{1}{\sqrt{p}} + \frac{1}{\sqrt{q}} + \frac{1}{\sqrt{T}} + \frac{\sqrt{q}}{p} + \sqrt{\frac{q}{Tp}} \right) \log^2(T) \log(p) \log^2(q) \right\},
\end{aligned}$$

where the last line used (S.31). Hence in particular,

$$\begin{aligned}
\mathbb{P}_{y,\alpha}(y_{\alpha,t} \leq \hat{q}_{x,\alpha}(\theta)) &= \mathbb{F}_{y,\alpha}(\hat{q}_{x,\alpha}(\theta)) \geq \mathbb{F}_{x,\alpha}(\hat{q}_{x,\alpha}(\theta)) - \sup_{c \in \mathbb{R}} |\mathbb{F}_{y,\alpha}(c) - \mathbb{F}_{x,\alpha}(c)| \\
&\geq \theta + O_P \left\{ \left(\frac{1}{\sqrt{p}} + \frac{1}{\sqrt{q}} + \frac{1}{\sqrt{T}} + \frac{\sqrt{q}}{p} + \sqrt{\frac{q}{Tp}} \right) \log^2(T) \log(p) \log^2(q) \right\},
\end{aligned}$$

which is the result we want. $\square$

Proof of Theorem 9. We only consider $\hat{k}_r$, i.e. result 1 in Theorem 9, since the proof concerning $\hat{k}_c$ is in parallel to that for $\hat{k}_r$. For $j \in [k_r]$, first consider

$$\begin{aligned}
\lambda_j \left(\frac{1}{T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t' \right) &= \lambda_j \left(\frac{1}{T} \sum_{t=1}^T \mathbf{A}_r \mathbf{F}_t \mathbf{A}_c' \mathbf{A}_c \mathbf{F}_t' \mathbf{A}_r' \right) = \lambda_j \left(\mathbf{A}_r' \mathbf{A}_r \cdot \frac{1}{T} \sum_{t=1}^T \mathbf{F}_t \mathbf{A}_c' \mathbf{A}_c \mathbf{F}_t' \right) \\
&\asymp_P \text{tr}(\mathbf{A}_c' \mathbf{A}_c) \cdot \lambda_j(\mathbf{A}_r' \mathbf{A}_r) = \|\mathbf{A}_c\|_F^2 \cdot \lambda_j(\mathbf{\Sigma}_{A,r}^{1/2} \mathbf{Z}_r \mathbf{\Sigma}_{A,r}^{1/2}) \asymp q^{\delta_{c,1}} \cdot \lambda_j(\mathbf{Z}_r) = p^{\delta_{r,j}} q^{\delta_{c,1}},
\end{aligned} \tag{S.33}$$

where in the second line, we used Assumption (F1) in the first step and Assumption (L1) in the second. For the second last step, we used Theorem 1 of Ostrowski (1959) on the eigenvalues of a congruent transformation $\mathbf{\Sigma}_{A,r}^{1/2} \mathbf{Z}_r \mathbf{\Sigma}_{A,r}^{1/2}$ of $\mathbf{Z}_r$, where we further used Assumption (L1) that all eigenvalues of $\mathbf{\Sigma}_{A,r}$ are bounded away from 0 and infinity.

Since we have $T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t' = T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t' + T^{-1} \sum_{t=1}^T \mathbf{R}_{r,t}$ from (S.15), it holds by Weyl's inequality that for $j \in [k_r]$,

$$\left| \lambda_j \left(\frac{1}{T} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t' \right) - \lambda_j \left(\frac{1}{T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t' \right) \right| \leq \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{R}_{r,t} \right\| = o_P(\omega_r), \tag{S.34}$$

where $\omega_r = p^{\delta_{r,k_r}} q^{\delta_{c,1}}$ as defined in Lemma 3, and the last equality used $\boldsymbol{\gamma}'(T^{-1} \sum_{t=1}^T \mathbf{R}_{r,t})\boldsymbol{\gamma} = o_P(\omega_r)$ for any unit vector $\boldsymbol{\gamma} \in \mathbb{R}^p$ from the proof of Lemma 3. With our choice of ξ_r , we also have

$$\xi_r / \omega_r \asymp p^{1-\delta_{r,k_r}} q^{1-\delta_{c,1}} [(Tq)^{-1/2} + p^{-1/2}] = o(1), \tag{S.35}$$

where we used Assumption (R1) in the last equality. Hence for $k_r > 1$, if $j \in [k_r - 1]$, using (S.34) and (S.35) we have

$$\begin{aligned}
&\frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t') + \xi_r}{\lambda_j(T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t') + \xi_r} \\
&\geq \frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t') + \xi_r - |\lambda_{j+1}(T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t') - \lambda_{j+1}(T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t')|}{\lambda_j(T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t') + \xi_r + |\lambda_j(T^{-1} \sum_{t=1}^T \hat{\mathbf{L}}_t \hat{\mathbf{L}}_t') - \lambda_j(T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t')|} \\
&= \frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t') + o_P(\omega_r)}{\lambda_j(T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t') + o_P(\omega_r)} = \frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t')}{\lambda_j(T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t')} (1 + o_P(1)) \asymp_P p^{\delta_{r,j+1} - \delta_{r,j}},
\end{aligned} \tag{S.36}$$

where the last line used (S.33). Moreover, for any $j \in [k_r - 1]$,

$$\begin{aligned} \frac{\lambda_{k_r+1}(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t') + \xi_r}{\lambda_{k_r}(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t') + \xi_r} &= \frac{\lambda_{k_r+1}(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t') + \xi_r}{\omega_r(1 + o_P(1))} = O_P\left\{\frac{\lambda_{k_r+1}(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t')}{\omega_r} + \frac{\xi_r}{\omega_r}\right\} \\ &= O_P\left\{O_P(pq[(Tq)^{-1/2} + p^{-1/2}]/\omega_r) + \xi_r/\omega_r\right\} = O_P(\xi_r/\omega_r) = o_P(p^{\delta_{r,j+1}-\delta_{r,j}}), \end{aligned} \quad (\text{S.37})$$

where we used (S.35) and the proof of Lemma 3 in the first equality, our choice of ξ_r in the second last, and the extra rate assumption in the statement of the theorem in the last. In the third equality, we used the following (which will be shown at the end of this proof),

$$\lambda_j\left(\frac{1}{T} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t'\right) = O_P(T^{-1/2}pq^{1/2} + p^{1/2}q), \quad j = k_r + 1, \dots, p. \quad (\text{S.38})$$

Hence for $j = k_r + 1, \dots, \lfloor p/2 \rfloor$ (true also for $k_r = 1$),

$$\frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t') + \xi_r}{\lambda_j(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t') + \xi_r} \geq \frac{\xi_r/\omega_r}{O_P(\xi_r/\omega_r) + \xi_r/\omega_r} \geq \frac{1}{C} \quad (\text{S.39})$$

in probability for some generic positive constant C , where we used again our choice of ξ_r and (S.38) in the first inequality. Combining (S.36), (S.37) and (S.39), we may conclude our proposed $\widehat{k}_r$ is consistent for k_r .

If $k_r = 1$, then from our choice of ξ_r , (S.37) becomes

$$\frac{\lambda_{k_r+1}(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t') + \xi_r}{\lambda_{k_r}(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t') + \xi_r} = O_P(\xi_r/\omega_r) = o_P(1).$$

Together with (S.39) which holds true for $k_r = 1$, we can also conclude $\widehat{k}_r$ is consistent for k_r .

It remains to show (S.38). To this end, from (S.33) and (S.34), the first k_r eigenvalues of $T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t'$ coincides with those of $T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t'$ asymptotically, so that the first k_r eigenvectors corresponding to $T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t'$ coincides with those for $T^{-1} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t'$ asymptotically as $T, p \rightarrow \infty$, which are necessarily in $\mathcal{N}^\perp := \text{Span}(\mathbf{Q}_r)$, the linear span of the columns of $\mathbf{Q}_r$. This means that the (k_r+1) -th largest eigenvalue of $T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t'$ and beyond will asymptotically have eigenvectors in $\mathcal{N}$, the orthogonal complement of $\mathcal{N}^\perp$. Then for any unit vectors $\gamma \in \mathcal{N}$, we have from (S.17) and Lemma 2 that

$$\gamma' \left(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t' \right) \gamma = \gamma' \left(\frac{1}{T} \sum_{t=1}^T \mathbf{R}_{r,t} \right) \gamma = O_P \left(T^{-1} \left\| \sum_{t=1}^T \mathbf{R}_{r,t} \right\|_F \right) = O_P(T^{-1/2}pq^{1/2} + p^{1/2}q),$$

which is equivalent to (S.38). This completes the proof of Theorem 9. $\square$

Proof of Theorem 10. We only consider the proof for $\widehat{r}$, since the proof for $\widehat{\ell}$ is similar. Under model (3.1), and (3.2) and (3.3) for $\widehat{\mu}_t$ and $\widehat{\alpha}_t$ respectively, we can write

$$\widehat{\alpha}_t = \alpha_t + q^{-1} \mathbf{E}_t \mathbf{1}_q - (pq)^{-1} \mathbf{1}_p \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q,$$

so that we can decompose

$$\begin{aligned}
T^{-1} \sum_{t=1}^T \hat{\alpha}_t \hat{\alpha}_t' &= T^{-1} \sum_{t=1}^T \alpha_t \alpha_t' + R, \quad \text{where } R := \sum_{i=1}^5 J_i, \quad \text{and} \\
J_1 &:= T^{-1} q^{-2} \sum_{t=1}^T \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t', & J_2 &:= T^{-1} \sum_{t=1}^T ((pq)^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q)^2 \mathbf{1}_p \mathbf{1}_p', \\
J_3 &:= T^{-1} q^{-1} \sum_{t=1}^T (\alpha_t \mathbf{1}_q' \mathbf{E}_t' + \mathbf{E}_t \mathbf{1}_q \alpha_t'), & J_4 &:= -T^{-1} \sum_{t=1}^T (pq)^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q (\mathbf{1}_p \alpha_t' + \alpha_t \mathbf{1}_p'), \\
J_5 &:= -T^{-1} q^{-1} \sum_{t=1}^T (pq)^{-1} \mathbf{1}_p' \mathbf{E}_t \mathbf{1}_q (\mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' + \mathbf{1}_p \mathbf{1}_q' \mathbf{E}_t').
\end{aligned} \tag{S.40}$$

Hereafter we write $\boldsymbol{\eta}_t := \boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\epsilon}_t$. Consider

$$\begin{aligned}
\|J_1\| &= q^{-2} \|T^{-1/2} (\mathbf{E}_1 \mathbf{1}_q, \dots, \mathbf{E}_T \mathbf{1}_q)\|^2 \\
&\leq 2q^{-2} \|T^{-1/2} (\mathbf{A}_{e,r} \mathbf{F}_{e,1} \mathbf{A}_{e,c}' \mathbf{1}_q, \dots, \mathbf{A}_{e,r} \mathbf{F}_{e,T} \mathbf{A}_{e,c}' \mathbf{1}_q)\|^2 + 2q^{-2} \|T^{-1/2} (\boldsymbol{\eta}_1 \mathbf{1}_q, \dots, \boldsymbol{\eta}_T \mathbf{1}_q)\|^2 \\
&=: J_{11} + J_{12}.
\end{aligned} \tag{S.41}$$

Since both $\mathbf{A}_{e,r}$ and $\mathbf{A}_{e,c}$ are (approximately) sparse in the L_1 sense by Assumption (E1), and $\mathbf{F}_{e,t}$ has finite second order moments by Assumption (E2), we have $J_{11} = O_P(q^{-2})$.

For J_{12} , rewrite $\boldsymbol{\eta}_t \mathbf{1}_q = \text{diag}^{1/2}((\boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\Sigma}_\epsilon) \mathbf{1}_q) \cdot \sum_{w \geq 0} a_{\epsilon,w} \mathbf{z}_{t-w}$, where

$$\mathbf{z}_t := \text{diag}^{-1/2}((\boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\Sigma}_\epsilon) \mathbf{1}_q) (\boldsymbol{\Sigma}_\epsilon \circ \mathbf{X}_{\epsilon,t}) \mathbf{1}_q.$$

Then $\{\mathbf{z}_t\}$ is a sequence of independent and identically distributed p -dimensional random vectors, with independent elements in each $\mathbf{z}_t$ having mean 0, unit variance and bounded fourth order moments for each $t \in [T]$ by Assumption (E1) and (E2). Also, Assumption (ME2) let the sequence $\{a_{\epsilon,w}\}_{w \in \mathbb{N} \cup \{0\}}$ satisfies Assumption 2.1 in Liu et al. (2015). Hence together with $p/T \rightarrow c_p \in (0, \infty)$,

$$\begin{aligned}
J_{12} &= 2q^{-2} \|T^{-1/2} (\boldsymbol{\eta}_1 \mathbf{1}_q, \dots, \boldsymbol{\eta}_T \mathbf{1}_q)\|^2 \\
&\leq 2q^{-2} \|\text{diag}((\boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\Sigma}_\epsilon) \mathbf{1}_q)\| \cdot \left\| T^{-1} \sum_{t=1}^T \left(\sum_{w \geq 0} a_{\epsilon,w} \mathbf{z}_{t-w} \right) \left(\sum_{w \geq 0} a_{\epsilon,w} \mathbf{z}_{t-w} \right)' \right\| \\
&= 2q^{-2} \cdot O(q) \cdot O_P(1) = O_P(q^{-1}),
\end{aligned}$$

where the last line follows from Theorem 2.1 of Liu et al. (2015). Hence together with the rate for J_{11} , we have

$$\|J_1\| = O_P(q^{-1}). \tag{S.42}$$

For $\|J_2\|$, note that

$$J_2 = p^{-2} \mathbf{1}_p \mathbf{1}_p' \cdot J_1 \cdot \mathbf{1}_p \mathbf{1}_p'$$

so that

$$\|J_2\| = O_P(p^{-2} \cdot p \cdot q^{-1} \cdot p) = O_P(q^{-1}). \tag{S.43}$$

Similarly, noting that $J_5 = p^{-1} J_1 \mathbf{1}_p \mathbf{1}_p' + p^{-1} \mathbf{1}_p \mathbf{1}_p' J_1$, we have

$$\|J_5\| = O_P(\|J_1\|) = O_P(q^{-1}). \tag{S.44}$$

For J_3 , we have

$$\begin{aligned}
\|J_3\| &\leq 2q^{-1}\|\boldsymbol{\alpha}\| \cdot \|T^{-1/2}(\mathbf{g}_1, \dots, \mathbf{g}_T)\| \cdot \|T^{-1/2}(\mathbf{E}_1 \mathbf{1}_q, \dots, \mathbf{E}_T \mathbf{1}_q)\| \\
&= 2q^{-1}\|\boldsymbol{\alpha}\| \cdot \|T^{-1/2}(\mathbf{g}_1, \dots, \mathbf{g}_T)\| \cdot q\|J_1\|^{1/2} \\
&= O(p^{\gamma_\alpha}) \cdot \|T^{-1/2}(\mathbf{g}_1, \dots, \mathbf{g}_T)\| \cdot O_P(q^{-1/2}) \\
&= O_P(p^{\gamma_\alpha} q^{-1/2}),
\end{aligned} \tag{S.45}$$

where we use Theorem 1 of Yao (2012) when $r/T \rightarrow c_r \in (0, 1)$ to conclude that $\|T^{-1/2}(\mathbf{g}_1, \dots, \mathbf{g}_T)\| = O_P(1)$. When r is finite, we use Theorem 2.2 of Bhansali et al. (2007), a CLT for quadratic forms of general linear processes, to conclude that (with Assumption 2.1 there satisfied at $d = 0$ from the tail condition of the sequence $\{a_{\epsilon, w}\}_{w \in \mathbb{N} \cup \{0\}}$ assumed in (ME1), and Assumption 2.2 satisfied with $\beta = 0$, $k_n = \eta_n(\lambda) = n^{-1}$)

$$\|T^{-1/2}(\mathbf{g}_1, \dots, \mathbf{g}_T)\| = \left(T^{-1} \sum_{t=1}^T \mathbf{g}_t' \mathbf{g}_t\right)^{1/2} \xrightarrow{\mathcal{P}} r^{1/2}.$$

For J_4 , it is not difficult to see that

$$\begin{aligned}
\|J_4\| &\leq 2\|\boldsymbol{\alpha}\| \cdot \|T^{-1/2}(\mathbf{g}_1, \dots, \mathbf{g}_T)\| \cdot \|T^{-1/2}((pq)^{-1}(\mathbf{1}_p' \mathbf{E}_1 \mathbf{1}_q) \mathbf{1}_p, \dots, (pq)^{-1}(\mathbf{1}_p' \mathbf{E}_T \mathbf{1}_q) \mathbf{1}_p)\| \\
&= 2\|\boldsymbol{\alpha}\| \cdot \|T^{-1/2}(\mathbf{g}_1, \dots, \mathbf{g}_T)\| \cdot \|J_2\|^{1/2} \\
&= O_P(p^{\gamma_\alpha} q^{-1/2}),
\end{aligned} \tag{S.46}$$

where we use the exact arguments before for proving the rate $\|T^{-1/2}(\mathbf{g}_1, \dots, \mathbf{g}_T)\| = O_P(1)$, and (S.43) for the rate $\|J_2\|^{1/2} = O_P(q^{-1/2})$.

Hence from (S.42) to (S.46), we have

$$\|R\| = O_P(p^{\gamma_\alpha} q^{-1/2}). \tag{S.47}$$

At the same time, for $j \in [r]$, using Theorem 1 and 4 of Wang and Zhang (1992), we have

$$\lambda_r \left(T^{-1} \sum_{t=1}^T \mathbf{g}_t' \mathbf{g}_t'\right) \lambda_j(\boldsymbol{\alpha}' \boldsymbol{\alpha}) \leq \lambda_j \left(T^{-1} \sum_{t=1}^T \boldsymbol{\alpha}_t \boldsymbol{\alpha}_t'\right) \leq \lambda_j(\boldsymbol{\alpha}' \boldsymbol{\alpha}) \lambda_1 \left(T^{-1} \sum_{t=1}^T \mathbf{g}_t' \mathbf{g}_t'\right).$$

But if $r/T \rightarrow c_r \in (0, 1)$, Theorem 1 of Yao (2012) implies that for $i \in [r]$, $\lambda_i(T^{-1} \sum_{t=1}^T \mathbf{g}_t' \mathbf{g}_t') \asymp_P 1$ which is bounded away from 0. If r is finite, similar argument used in (S.45) using Theorem 2.2 of Bhansali et al. (2007) results in $T^{-1} \sum_{t=1}^T \mathbf{g}_t' \mathbf{g}_t' \xrightarrow{\mathcal{P}} \mathbf{I}_r$, so that we still have $\lambda_i(T^{-1} \sum_{t=1}^T \mathbf{g}_t' \mathbf{g}_t') \asymp_P 1$ for $i \in [r]$. Hence, we have for $j \in [r]$,

$$\lambda_j \left(T^{-1} \sum_{t=1}^T \boldsymbol{\alpha}_t \boldsymbol{\alpha}_t'\right) \asymp_P \lambda_j(\boldsymbol{\alpha}' \boldsymbol{\alpha}) \asymp p^{2\gamma_\alpha}. \tag{S.48}$$

Suppose we also have (to be proved at the end of this proof) for $j = r+1, \dots, p$,

$$\lambda_j \left(T^{-1} \sum_{t=1}^T \hat{\boldsymbol{\alpha}}_t \hat{\boldsymbol{\alpha}}_t'\right) = O_P(\|R\|) = O_P(p^{\gamma_\alpha} q^{-1/2}). \tag{S.49}$$

Then for $j \in [r-1]$ (if $r > 1$),

$$\frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \hat{\boldsymbol{\alpha}}_t \hat{\boldsymbol{\alpha}}_t') + q^{-1/2}}{\lambda_j(T^{-1} \sum_{t=1}^T \hat{\boldsymbol{\alpha}}_t \hat{\boldsymbol{\alpha}}_t') + q^{-1/2}} \geq \frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \boldsymbol{\alpha}_t \boldsymbol{\alpha}_t') - \|R\| + q^{-1/2}}{\lambda_j(T^{-1} \sum_{t=1}^T \boldsymbol{\alpha}_t \boldsymbol{\alpha}_t') + \|R\| + q^{-1/2}} \asymp_P \frac{p^{2\gamma_\alpha}(1 + o_P(1))}{p^{2\gamma_\alpha}(1 + o_P(1))} = 1.$$

And for $r \geq 1$,

$$\frac{\lambda_{r+1}(T^{-1} \sum_{t=1}^T \hat{\alpha}_t \hat{\alpha}_t' + q^{-1/2})}{\lambda_r(T^{-1} \sum_{t=1}^T \hat{\alpha}_t \hat{\alpha}_t' + q^{-1/2})} = \frac{O_P(p^{\gamma_\alpha} q^{-1/2} + q^{-1/2})}{p^{2\gamma_\alpha}(1 + o_P(1))} = O_P(p^{-\gamma_\alpha} q^{-1/2}).$$

For $j = r + 1, \dots, p$,

$$\frac{\lambda_{j+1}(T^{-1} \sum_{t=1}^T \hat{\alpha}_t \hat{\alpha}_t' + q^{-1/2})}{\lambda_j(T^{-1} \sum_{t=1}^T \hat{\alpha}_t \hat{\alpha}_t' + q^{-1/2})} \geq \frac{q^{-1/2}}{O_P(p^{\gamma_\alpha} q^{-1/2}) + q^{-1/2}} \succ_P p^{-\gamma_\alpha} \succ_P p^{-\gamma_\alpha} q^{-1/2},$$

where $a \succ_P b$ means $b = O_P(a)$. Combining the above, we have the consistency of $\hat{r}$ for r as $\min(p, q, T) \rightarrow \infty$. It remains to show (S.49).

In fact, (S.47) and (S.48), together with the decomposition (S.40), show that for the r largest eigenvalues of $T^{-1} \sum_{t=1}^T \hat{\alpha}_t \hat{\alpha}_t'$, the corresponding eigenvectors are asymptotically in the span of the columns of α , $\text{Span}(\alpha)$, so that these eigenvalues are of order $p^{2\gamma_\alpha}$ in probability. At the same time, for the j -th largest eigenvalue where $j = r + 1, \dots, p$, since $\text{Span}(\alpha)$ has rank r only, the corresponding eigenvector must be asymptotically belongs to the orthogonal space of $\text{Span}(\alpha)$, or some of the r largest eigenvalues can in fact be larger by absorbing similar directions into their eigenvectors, contradicting that they are already the r largest eigenvalues. This means that for $j = r + 1, \dots, p$,

$$\lambda_j \left(T^{-1} \sum_{t=1}^T \hat{\alpha}_t \hat{\alpha}_t' \right) = \lambda_j \left(T^{-1} \sum_{t=1}^T \alpha_t \alpha_t' + R \right) \rightarrow \lambda_j(R) = O_P(\|R\|) = O_P(p^{\gamma_\alpha} q^{-1/2}),$$

which is (S.49). This completes the proof of the theorem. $\square$

2 Lemmas and proofs

As we have the same factor structure as Cen and Lam (2025), we list the following Lemma 1 for further use and refer readers to Cen and Lam (2025) for the proof in details.

Lemma 1 *Let Assumptions (F1), (E1) and (E2) hold. Then*

1. *(Weak correlation of noise $\mathbf{E}_t$ across different rows, columns and times) there exists some positive constant $C < \infty$ so that for any $t \in [T], i, j \in [p], h \in [q]$,*

$$\begin{aligned} \sum_{k=1}^p \sum_{l=1}^q \left| \mathbb{E}[E_{t,ih} E_{t,kl}] \right| &\leq C, \\ \sum_{l=1}^q \sum_{s=1}^T \left| \text{cov}(E_{t,ih} E_{t,jh}, E_{s,il} E_{s,jl}) \right| &\leq C. \end{aligned}$$

2. *(Weak dependence between factor $\mathbf{F}_t$ and noise $\mathbf{E}_t$) there exists some positive constant $C < \infty$ so that for any $j \in [p], i \in [q]$, and any deterministic vectors $\mathbf{u} \in \mathbb{R}^{k_r}$ and $\mathbf{v} \in \mathbb{R}^{k_c}$ with constant magnitudes,*

$$\mathbb{E} \left(\frac{1}{(qT)^{1/2}} \sum_{h=1}^q \sum_{t=1}^T E_{t,jh} \mathbf{u}' \mathbf{F}_t \mathbf{v} \right)^2 \leq C, \quad \mathbb{E} \left(\frac{1}{(pT)^{1/2}} \sum_{h=1}^p \sum_{t=1}^T E_{t,hi} \mathbf{v}' \mathbf{F}_t' \mathbf{u} \right)^2 \leq C.$$

3. *(Further results on factor $\mathbf{F}_t$) for any $t \in [T]$, all elements in $\mathbf{F}_t$ are independent of each other, with mean 0 and unit variance. Moreover,*

$$\frac{1}{T} \sum_{t=1}^T \mathbf{F}_t \mathbf{F}_t' \xrightarrow{p} \Sigma_r := k_c \mathbf{I}_{k_r}, \quad \frac{1}{T} \sum_{t=1}^T \mathbf{F}_t' \mathbf{F}_t \xrightarrow{p} \Sigma_c := k_r \mathbf{I}_{k_c},$$

with the number of factors k_r and k_c fixed as $\min\{T, p, q\} \rightarrow \infty$.

Lemma 2 (Bounding $\sum_{t=1}^T \mathbf{R}_{r,t}$) Under Assumptions (F1), (L1), (E1) and (E2), it holds that

$$\left\| \sum_{t=1}^T \mathbf{C}_t \mathbf{E}_t' \right\|_F^2 = O_P(Tp^{1+\delta_{r,1}}q), \quad (\text{S.1})$$

$$\left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{E}_t' \right\|_F^2 = O_P(Tp^2q + T^2pq^2), \quad (\text{S.2})$$

$$\left\| \sum_{t=1}^T \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' \right\|_F^2 = O_P(Tp^3q^2 + T^2p^2q^2), \quad (\text{S.3})$$

$$\left\| \sum_{t=1}^T \mathbf{C}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 = O_P(Tp^{3+\delta_{r,1}}q), \quad (\text{S.4})$$

$$\left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 = O_P(Tp^4q + T^2p^3q^2), \quad (\text{S.5})$$

$$\left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \right\|_F^2 = O_P(Tp^2q^2 + T^2pq^2), \quad (\text{S.6})$$

$$\left\| \sum_{t=1}^T (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p)^2 \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 = O_P(T^2p^4q^2), \quad (\text{S.7})$$

$$\left\| \sum_{t=1}^T \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 = O_P(Tp^6q + T^2p^5q^2). \quad (\text{S.8})$$

Thus, with $\mathbf{R}_{r,t}$ defined in (S.15), we have

$$\left\| \sum_{t=1}^T \mathbf{R}_{r,t} \right\|_F^2 = O_P(Tp^2q + T^2pq^2).$$

Proof of Lemma 2. Using $\mathbf{C}_t = \mathbf{A}_r \mathbf{F}_t \mathbf{A}_c'$, we have (S.1) holds as follows,

$$\begin{aligned} \left\| \sum_{t=1}^T \mathbf{C}_t \mathbf{E}_t' \right\|_F^2 &= \left\| \sum_{t=1}^T \mathbf{A}_r \mathbf{F}_t \mathbf{A}_c' \mathbf{E}_t' \right\|_F^2 = \sum_{i=1}^p \sum_{l=1}^p \left(\sum_{t=1}^T \mathbf{A}_{r,i}' \mathbf{F}_t \mathbf{A}_{c,l}' \mathbf{E}_{t,l}' \right)^2 \\ &= \sum_{i=1}^p \|\mathbf{A}_{r,i}\|^2 \cdot \sum_{l=1}^p \left(\sum_{h=1}^q \sum_{t=1}^T E_{t,hl} \frac{1}{\|\mathbf{A}_{r,i}\|} \mathbf{A}_{r,i}' \mathbf{F}_t \mathbf{A}_{c,h}' \right)^2 = O_P(Tp^{1+\delta_{r,1}}q), \end{aligned}$$

where the last equality is from Assumption (L1) and Lemma 1.

To show (S.2), first notice from Assumption (E1),

$$E_{t,ij} = \mathbf{A}_{e,r,i}' \mathbf{F}_{e,t} \mathbf{A}_{e,c,j} + \Sigma_{\epsilon,ij} \epsilon_{t,ij}.$$

With Assumption (E2), we have

$$\text{cov}(E_{t,ij}, E_{t,kj}) = \mathbf{A}_{e,r,i}' \mathbf{A}_{e,r,k} \|\mathbf{A}_{e,c,j}\|^2 + \Sigma_{\epsilon,ij}^2 \mathbb{1}_{\{i=k\}},$$

and hence using Lemma 1,

$$\begin{aligned} \mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{E}_t' \right\|_F^2 \right] &= \sum_{i=1}^p \sum_{k=1}^p \mathbb{E} \left[\left(\sum_{t=1}^T \sum_{j=1}^q E_{t,ij} E_{t,kj} \right)^2 \right] \\ &= \sum_{i=1}^p \sum_{k=1}^p \left[\sum_{t=1}^T \sum_{j=1}^q \sum_{s=1}^T \sum_{l=1}^q \text{cov}(E_{t,ij} E_{t,kj}, E_{s,il} E_{s,kl}) + \left(\sum_{t=1}^T \sum_{j=1}^q \mathbb{E}[E_{t,ij} E_{t,kj}] \right)^2 \right] \\ &= O(Tp^2q) + \sum_{i=1}^p \sum_{k=1}^p O \left(T \cdot \mathbf{A}_{e,r,i}' \mathbf{A}_{e,r,k} \|\mathbf{A}_{e,c}\|_F^2 + Tq \cdot \mathbb{1}_{\{i=k\}} \right)^2 = O(Tp^2q + T^2pq^2). \end{aligned}$$

For (S.3), consider first

$$\begin{aligned}
& \sum_{t=1}^T \sum_{s=1}^T \text{cov}(E_{t,ij} E_{t,kh}, E_{s,lm} E_{s,kn}) \\
&= \sum_{t=1}^T \sum_{s=1}^T \text{cov} \left[(\mathbf{A}'_{e,r,i} \mathbf{F}_{e,t} \mathbf{A}_{e,c,j} + \Sigma_{\epsilon,ij} \epsilon_{t,ij}) (\mathbf{A}'_{e,r,k} \mathbf{F}_{e,t} \mathbf{A}_{e,c,h} + \Sigma_{\epsilon,kh} \epsilon_{t,kh}), \right. \\
& \quad \left. (\mathbf{A}'_{e,r,l} \mathbf{F}_{e,s} \mathbf{A}_{e,c,m} + \Sigma_{\epsilon,lm} \epsilon_{s,lm}) (\mathbf{A}'_{e,r,k} \mathbf{F}_{e,s} \mathbf{A}_{e,c,n} + \Sigma_{\epsilon,kn} \epsilon_{s,kn}) \right] \\
&= \sum_{t=1}^T \sum_{s=1}^T \text{cov} \left[\mathbf{A}'_{e,r,i} \mathbf{F}_{e,t} \mathbf{A}_{e,c,j} \mathbf{A}'_{e,r,k} \mathbf{F}_{e,t} \mathbf{A}_{e,c,h}, \mathbf{A}'_{e,r,l} \mathbf{F}_{e,s} \mathbf{A}_{e,c,m} \mathbf{A}'_{e,r,k} \mathbf{F}_{e,s} \mathbf{A}_{e,c,n} \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,i} \mathbf{F}_{e,t} \mathbf{A}_{e,c,j} \mathbf{A}'_{e,r,l} \mathbf{F}_{e,s} \mathbf{A}_{e,c,m} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,kh} \epsilon_{t,kh} \Sigma_{\epsilon,kn} \epsilon_{s,kn} \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,i} \mathbf{F}_{e,t} \mathbf{A}_{e,c,j} \mathbf{A}'_{e,r,k} \mathbf{F}_{e,s} \mathbf{A}_{e,c,n} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,kh} \epsilon_{t,kh} \Sigma_{\epsilon,lm} \epsilon_{s,lm} \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,k} \mathbf{F}_{e,t} \mathbf{A}_{e,c,h} \mathbf{A}'_{e,r,l} \mathbf{F}_{e,s} \mathbf{A}_{e,c,m} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,ij} \epsilon_{t,ij} \Sigma_{\epsilon,kn} \epsilon_{s,kn} \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,k} \mathbf{F}_{e,t} \mathbf{A}_{e,c,h} \mathbf{A}'_{e,r,k} \mathbf{F}_{e,s} \mathbf{A}_{e,c,n} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,ij} \epsilon_{t,ij} \Sigma_{\epsilon,lm} \epsilon_{s,lm} \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \text{cov} \left[\Sigma_{\epsilon,ij} \epsilon_{t,ij} \Sigma_{\epsilon,kh} \epsilon_{t,kh}, \Sigma_{\epsilon,lm} \epsilon_{s,lm} \Sigma_{\epsilon,kn} \epsilon_{s,kn} \right] \\
&= \sum_{t=1}^T \sum_{s=1}^T \text{cov} \left[\mathbf{A}'_{e,r,i} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,j} \mathbf{A}'_{e,r,k} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,h}, \right. \\
& \quad \mathbf{A}'_{e,r,l} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,m} \mathbf{A}'_{e,r,k} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,n} \left. \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,i} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,j} \mathbf{A}'_{e,r,l} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,m} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,kh} \epsilon_{t,kh} \Sigma_{\epsilon,kn} \epsilon_{s,kn} \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,i} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,j} \mathbf{A}'_{e,r,k} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,n} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,kh} \epsilon_{t,kh} \Sigma_{\epsilon,lm} \epsilon_{s,lm} \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,k} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,h} \mathbf{A}'_{e,r,l} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,m} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,ij} \epsilon_{t,ij} \Sigma_{\epsilon,kn} \epsilon_{s,kn} \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,k} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,h} \mathbf{A}'_{e,r,k} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,n} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,ij} \epsilon_{t,ij} \Sigma_{\epsilon,lm} \epsilon_{s,lm} \right] \\
&+ \sum_{t=1}^T \sum_{s=1}^T \text{cov} \left[\Sigma_{\epsilon,ij} \epsilon_{t,ij} \Sigma_{\epsilon,kh} \epsilon_{t,kh}, \Sigma_{\epsilon,lm} \epsilon_{s,lm} \Sigma_{\epsilon,kn} \epsilon_{s,kn} \right].
\end{aligned} \tag{S.9}$$

Consider the six terms in the last equality above, we have the first term as

$$\begin{aligned}
& \sum_{t=1}^T \sum_{s=1}^T \text{cov} \left[\mathbf{A}'_{e,r,i} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,k} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,h}, \right. \\
& \left. \mathbf{A}'_{e,r,l} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,m} \cdot \mathbf{A}'_{e,r,k} \cdot \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,n} \right] \\
&= \sum_{t=1}^T \sum_{w \geq 0} a_{e,w}^2 \cdot \mathbb{E} \left[\mathbf{A}'_{e,r,i} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,n} \right] \cdot \mathbb{E} \left[\mathbf{A}'_{e,r,k} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h} \cdot \mathbf{A}'_{e,r,l} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,m} \right] \\
&+ \sum_{t=1}^T \sum_{w \geq 0} a_{e,w}^2 \cdot \mathbb{E} \left[\mathbf{A}'_{e,r,i} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,l} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,m} \right] \cdot \mathbb{E} \left[\mathbf{A}'_{e,r,k} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,n} \right] \\
&+ \sum_{t=1}^T \sum_{w \geq 0} a_{e,w}^4 \cdot \mathbb{E} \left[\mathbf{A}'_{e,r,i} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h} \cdot \mathbf{A}'_{e,r,l} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,m} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,n} \right] \\
&- \sum_{t=1}^T \sum_{w \geq 0} a_{e,w}^4 \cdot \mathbb{E} \left[\mathbf{A}'_{e,r,i} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,h} \right] \cdot \mathbb{E} \left[\mathbf{A}'_{e,r,l} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,m} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{X}_{e,t-w} \mathbf{A}_{e,c,n} \right] \\
&= \sum_{t=1}^T \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,i} \cdot \mathbf{A}_{e,c,n} \cdot \mathbf{A}'_{e,r,l} \cdot \mathbf{A}_{e,c,h} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,m} \\
&+ \sum_{t=1}^T \mathbf{A}'_{e,r,l} \cdot \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,i} \cdot \mathbf{A}_{e,c,m} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,h} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,n} \\
&+ O(1) \cdot \sum_{t=1}^T \sum_{w \geq 0} a_{e,w}^4 \cdot \|\mathbf{A}_{e,r,k}\|^2 \cdot \|\mathbf{A}_{e,r,i}\| \cdot \|\mathbf{A}_{e,c,j}\| \cdot \|\mathbf{A}_{e,c,h}\| \cdot \|\mathbf{A}_{e,r,l}\| \cdot \|\mathbf{A}_{e,c,m}\| \cdot \|\mathbf{A}_{e,c,n}\| \\
&- \sum_{t=1}^T \sum_{w \geq 0} a_{e,w}^4 \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,i} \cdot \mathbf{A}_{e,c,h} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,m} \cdot \mathbf{A}'_{e,r,l} \cdot \mathbf{A}_{e,c,n},
\end{aligned} \tag{S.10}$$

where we used (E2) in the last equality that each entry in $\{\mathbf{X}_{e,t}\}$ is independent with uniformly bounded

fourth moment. Similarly, the remaining terms in last equality of (S.9) are

$$\begin{aligned} & \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,i} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,j} \mathbf{A}'_{e,r,l} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,m} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,kh} \epsilon_{t,kh} \Sigma_{\epsilon,kn} \epsilon_{s,kn} \right] \\ &= \sum_{t=1}^T \mathbf{A}'_{e,r,l} \cdot \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,i} \cdot \mathbf{A}_{e,c,m} \cdot \Sigma_{\epsilon,kh} \Sigma_{\epsilon,kn} \cdot \mathbb{1}_{\{h=n\}}, \end{aligned} \quad (\text{S.11})$$

$$\begin{aligned} & \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,i} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,j} \mathbf{A}'_{e,r,k} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,n} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,kh} \epsilon_{t,kh} \Sigma_{\epsilon,lm} \epsilon_{s,lm} \right] \\ &= \sum_{t=1}^T \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,j} \cdot \mathbf{A}'_{e,r,i} \cdot \mathbf{A}_{e,c,n} \cdot \Sigma_{\epsilon,kh} \Sigma_{\epsilon,lm} \cdot \mathbb{1}_{\{k=l\}} \mathbb{1}_{\{h=m\}}, \end{aligned} \quad (\text{S.12})$$

$$\begin{aligned} & \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,k} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,h} \mathbf{A}'_{e,r,l} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,m} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,ij} \epsilon_{t,ij} \Sigma_{\epsilon,kn} \epsilon_{s,kn} \right] \\ &= \sum_{t=1}^T \mathbf{A}'_{e,r,l} \cdot \mathbf{A}_{e,c,h} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,m} \cdot \Sigma_{\epsilon,ij} \Sigma_{\epsilon,kn} \cdot \mathbb{1}_{\{i=k\}} \mathbb{1}_{\{j=n\}}, \end{aligned} \quad (\text{S.13})$$

$$\begin{aligned} & \sum_{t=1}^T \sum_{s=1}^T \mathbb{E} \left[\mathbf{A}'_{e,r,k} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,h} \mathbf{A}'_{e,r,k} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,s-w} \right) \mathbf{A}_{e,c,n} \right] \cdot \mathbb{E} \left[\Sigma_{\epsilon,ij} \epsilon_{t,ij} \Sigma_{\epsilon,lm} \epsilon_{s,lm} \right] \\ &= \sum_{t=1}^T \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,h} \cdot \mathbf{A}'_{e,r,k} \cdot \mathbf{A}_{e,c,n} \cdot \Sigma_{\epsilon,ij} \Sigma_{\epsilon,lm} \cdot \mathbb{1}_{\{i=l\}} \mathbb{1}_{\{j=m\}}, \end{aligned} \quad (\text{S.14})$$

$$\begin{aligned} & \sum_{t=1}^T \sum_{s=1}^T \text{cov} \left[\Sigma_{\epsilon,ij} \epsilon_{t,ij} \Sigma_{\epsilon,kh} \epsilon_{t,kh}, \Sigma_{\epsilon,lm} \epsilon_{s,lm} \Sigma_{\epsilon,kn} \epsilon_{s,kn} \right] \\ &= O(T) \cdot \left(\mathbb{1}_{\{i=l=k\}} \mathbb{1}_{\{j=h=m=n\}} + \mathbb{1}_{\{i=l\}} \mathbb{1}_{\{j=m\}} \mathbb{1}_{\{h=n\}} + \mathbb{1}_{\{i=l=k\}} \mathbb{1}_{\{j=n\}} \mathbb{1}_{\{h=m\}} \right). \end{aligned} \quad (\text{S.15})$$

Using (S.10), (S.11), (S.12), (S.13), (S.14) and (S.15), we arrive at an expression for (S.9). Thus, (S.3) can be obtained as

$$\begin{aligned} & \mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{1}'_q \mathbf{E}'_t \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}'_p \right\|_F^2 \right] = p \sum_{k=1}^p \mathbb{E} \left\{ \left[\sum_{t=1}^T \left(\sum_{i=1}^p \sum_{j=1}^q E_{t,ij} \right) \sum_{h=1}^q E_{t,kh} \right]^2 \right\} \\ &= p \sum_{k=1}^p \left[\sum_{t=1}^T \sum_{i=1}^p \sum_{j=1}^q \sum_{h=1}^q \sum_{s=1}^T \sum_{l=1}^p \sum_{m=1}^q \sum_{n=1}^q \text{cov}(E_{t,ij} E_{t,kh}, E_{s,lm} E_{s,kn}) + \left(\sum_{t=1}^T \sum_{i=1}^p \sum_{j=1}^q \sum_{h=1}^q \mathbb{E}[E_{t,ij} E_{t,kh}] \right)^2 \right] \\ &= O(Tp^3q^2) + p \sum_{k=1}^p \left[\sum_{t=1}^T \sum_{i=1}^p \sum_{j=1}^q \sum_{h=1}^q (\mathbf{A}'_{e,r,i} \cdot \mathbf{A}_{e,r,k} \cdot \mathbf{A}'_{e,c,j} \cdot \mathbf{A}_{e,c,h} + \Sigma_{\epsilon,ij}^2 \mathbb{1}_{\{i=k\}} \mathbb{1}_{\{j=h\}}) \right]^2 \\ &= O(Tp^3q^2 + T^2p^2q^2). \end{aligned} \quad (\text{S.16})$$

By (S.1) and (S.2), we can obtain (S.4) and (S.5), respectively as follows,

$$\begin{aligned} \left\| \sum_{t=1}^T \mathbf{C}_t \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_p \right\|_F^2 &\leq \left\| \sum_{t=1}^T \mathbf{C}_t \mathbf{E}'_t \right\|_F^2 \cdot \|\mathbf{1}_p \mathbf{1}'_p\|_F^2 = O_P(Tp^{3+\delta_{r,1}}q), \\ \left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{E}'_t \mathbf{1}_p \mathbf{1}'_p \right\|_F^2 &\leq \left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{E}'_t \right\|_F^2 \cdot \|\mathbf{1}_p \mathbf{1}'_p\|_F^2 = O_P(Tp^4q + T^2p^3q^2). \end{aligned}$$

Similar to the proof of (S.3), we can show (S.6) by

$$\begin{aligned}
& \mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \right\|_F^2 \right] = \sum_{i=1}^p \sum_{k=1}^p \mathbb{E} \left\{ \left[\sum_{t=1}^T \left(\sum_{j=1}^q E_{t,ij} \right) \left(\sum_{h=1}^q E_{t,kh} \right) \right]^2 \right\} \\
&= \sum_{i=1}^p \sum_{k=1}^p \left[\sum_{t=1}^T \sum_{j=1}^q \sum_{h=1}^q \sum_{s=1}^T \sum_{m=1}^q \sum_{n=1}^q \text{cov}(E_{t,ij} E_{t,kh}, E_{s,im} E_{s,kn}) + \left(\sum_{t=1}^T \sum_{j=1}^q \sum_{h=1}^q \mathbb{E}[E_{t,ij} E_{t,kh}] \right)^2 \right] \\
&= O(Tp^2q^2) + \sum_{i=1}^p \sum_{k=1}^p \left[\sum_{t=1}^T \sum_{j=1}^q \sum_{h=1}^q (\mathbf{A}'_{e,r,i} \cdot \mathbf{A}_{e,r,k} \cdot \mathbf{A}'_{e,c,j} \cdot \mathbf{A}_{e,c,h} + \Sigma_{\epsilon,ij}^2 \mathbb{1}_{\{i=k\}} \mathbb{1}_{\{j=h\}}) \right]^2 \\
&= O(Tp^2q^2 + T^2pq^2).
\end{aligned}$$

From (S.2), we can obtain (S.7) such that

$$\left\| \sum_{t=1}^T (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p)^2 \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 = \left[\sum_{t=1}^T (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p)^2 \right]^2 \cdot \left\| \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 = O_P(T^2p^4q^2).$$

Lastly, from (S.2) we have

$$\left\| \sum_{t=1}^T \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 \leq \left\| \mathbf{1}_p \right\|^2 \cdot \left\| \mathbf{1}_p \right\|^2 \cdot \left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{E}_t' \right\|_F^2 \cdot \left\| \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 = O_P(Tp^6q + T^2p^5q^2).$$

From (S.15), we have

$$\begin{aligned}
\left\| \sum_{t=1}^T \mathbf{R}_{r,t} \right\|_F^2 &= O_P \left(\left\| \sum_{t=1}^T \mathbf{C}_t \mathbf{E}_t' \right\|_F^2 + \left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{E}_t' \right\|_F^2 + (pq)^{-2} \left\| \sum_{t=1}^T \mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p \mathbf{E}_t \mathbf{1}_q \mathbf{1}_p' \right\|_F^2 \right. \\
&\quad + p^{-2} \left\| \sum_{t=1}^T \mathbf{C}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 + p^{-2} \left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 + p^{-2} \left\| \sum_{t=1}^T \mathbf{E}_t \mathbf{1}_q \mathbf{1}_q' \mathbf{E}_t' \right\|_F^2 \\
&\quad \left. + (pq)^{-2} p^{-2} \left\| \sum_{t=1}^T (\mathbf{1}_q' \mathbf{E}_t' \mathbf{1}_p)^2 \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 + p^{-4} \left\| \sum_{t=1}^T \mathbf{1}_p' \mathbf{E}_t \mathbf{E}_t' \mathbf{1}_p \mathbf{1}_p \mathbf{1}_p' \right\|_F^2 \right) \\
&= O_P(Tp^2q + T^2pq^2),
\end{aligned}$$

which completes the proof of Lemma 2. $\square$

Lemma 3 *Let Assumptions (M1), (F1), (L1), (E1), (E2) and (R1) hold. Define $\omega_r := p^{\delta_r, k_r} q^{\delta_c, 1}$ and $\omega_c := q^{\delta_c, k_c} p^{\delta_r, 1}$. We have*

$$\|\widehat{\mathbf{D}}_r^{-1}\|_F = O_P(\omega_r^{-1}), \quad \|\widehat{\mathbf{D}}_c^{-1}\|_F = O_P(\omega_c^{-1}).$$

Proof of Lemma 3. It suffices to show the bound of $\|\widehat{\mathbf{D}}_r^{-1}\|_F$, since that of $\|\widehat{\mathbf{D}}_c^{-1}\|_F$ would be similar. First, we bound the term $\|\widehat{\mathbf{D}}_r^{-1}\|_F^2$ by finding the lower bound of $\lambda_{k_r}(\widehat{\mathbf{D}}_r)$. To do this, consider the decomposition

$$\frac{1}{T} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t' = \frac{1}{T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t' + \frac{1}{T} \sum_{t=1}^T \mathbf{R}_{r,t}, \quad (\text{S.17})$$

so that for a unit vector $\boldsymbol{\gamma} \in \mathbb{R}^p$, we can define

$$\begin{aligned} S_r(\boldsymbol{\gamma}) &:= \frac{1}{\omega_r} \boldsymbol{\gamma}' \left(\frac{1}{T} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t' \right) \boldsymbol{\gamma} =: S_r^*(\boldsymbol{\gamma}) + \widetilde{S}_r(\boldsymbol{\gamma}), \quad \text{with} \\ S_r^*(\boldsymbol{\gamma}) &:= \frac{1}{\omega_r} \boldsymbol{\gamma}' \left(\frac{1}{T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t' \right) \boldsymbol{\gamma}, \quad \widetilde{S}_r(\boldsymbol{\gamma}) := \frac{1}{\omega_r} \boldsymbol{\gamma}' \left(\frac{1}{T} \sum_{t=1}^T \mathbf{R}_{r,t} \right) \boldsymbol{\gamma}. \end{aligned}$$

Since $\|\boldsymbol{\gamma}\| = 1$, we have by Lemma 2,

$$|\widetilde{S}_r(\boldsymbol{\gamma})|^2 \leq \frac{1}{\omega_r^2 T^2} \left\| \sum_{t=1}^T \mathbf{R}_{r,t} \right\|_F^2 = O_P \left(T^{-1} p^{2(1-\delta_r, k_r)} q^{1-2\delta_c, 1} + p^{1-2\delta_r, k_r} q^{2(1-\delta_c, 1)} \right) = o_P(1),$$

where the last equality used Assumption (R1). Next, with Assumption (F1), consider

$$\begin{aligned} \lambda_{k_r} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t' \right) &= \lambda_{k_r} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{A}_r \mathbf{F}_t \mathbf{A}_c' \mathbf{A}_c \mathbf{F}_t' \mathbf{A}_r' \right) \geq \lambda_{k_r}(\mathbf{A}_r' \mathbf{A}_r) \cdot \lambda_{k_r} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{F}_t \mathbf{A}_c' \mathbf{A}_c \mathbf{F}_t' \right) \\ &\asymp_P p^{\delta_r, k_r} \cdot \lambda_{k_r}(\text{tr}(\mathbf{A}_c' \mathbf{A}_c) \boldsymbol{\Sigma}_r) \asymp_P p^{\delta_r, k_r} q^{\delta_c, 1} = \omega_r. \end{aligned}$$

With this, going back to the decomposition (S.17),

$$\omega_r^{-1} \lambda_{k_r}(\widehat{\mathbf{D}}_r) = \omega_r^{-1} \lambda_{k_r} \left(\frac{1}{T} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}_t' \right) \geq \omega_r^{-1} \lambda_{k_r} \left(\frac{1}{T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}_t' \right) - \sup_{\|\boldsymbol{\gamma}\|=1} |\widetilde{S}_r(\boldsymbol{\gamma})| \asymp_P 1,$$

and hence finally $\|\widehat{\mathbf{D}}_r^{-1}\|_F = O_P(\lambda_{k_r}^{-1}(\widehat{\mathbf{D}}_r)) = O_P(\omega_r^{-1})$, which completes the proof of Lemma 3. $\square$

Lemma 4 (*Limit of $\widehat{\mathbf{D}}_r$ and $\widehat{\mathbf{D}}_c$*) Let Assumptions (F1), (L1), (E1), (E2) and (R1) hold. With $\widehat{\mathbf{D}}_r, \widehat{\mathbf{D}}_c$ and ω_r, ω_c defined in Lemma 3, we have

$$\begin{aligned}\omega_r^{-1} \widehat{\mathbf{D}}_r &\xrightarrow{P} \omega_r^{-1} \mathbf{D}_r := \omega_r^{-1} \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \text{diag}\{\lambda_j(\mathbf{A}'_r \mathbf{A}_r) \mid j \in [k_r]\}, \\ \omega_c^{-1} \widehat{\mathbf{D}}_c &\xrightarrow{P} \omega_c^{-1} \mathbf{D}_c := \omega_c^{-1} \text{tr}(\mathbf{A}'_r \mathbf{A}_r) \cdot \text{diag}\{\lambda_j(\mathbf{A}'_c \mathbf{A}_c) \mid j \in [k_c]\}.\end{aligned}$$

Proof of Lemma 4. It suffices to show the limit of $\widehat{\mathbf{D}}_r$, as the proof for $\widehat{\mathbf{D}}_c$ will be similar. We first show

$$\frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}'_t \xrightarrow{P} \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} \mathbf{A}_r \mathbf{A}'_r, \quad (\text{S.18})$$

where $\lambda_{k_r}(\text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} \mathbf{A}_r \mathbf{A}'_r) \asymp 1$. By Assumption (F1), we have

$$\mathbb{E}\left(\frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}'_t\right) = \frac{1}{\omega_r T} \sum_{t=1}^T \mathbb{E}[\mathbf{A}_r \mathbf{F}_t \mathbf{A}'_c \mathbf{A}_c \mathbf{F}'_t \mathbf{A}'_r] = \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} \mathbf{A}_r \mathbf{A}'_r,$$

where we used the independence structure among elements in $\mathbf{F}_t$. Furthermore, for any $i, j \in [p]$,

$$\begin{aligned}\text{Var}\left[\left(\frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}'_t\right)_{ij}\right] &= \frac{1}{\omega_r^2 T^2} \sum_{t=1}^T \sum_{s=1}^T \text{cov}\left(\mathbf{A}'_{r,i} \cdot \left(\sum_{w \geq 0} a_{f,w} \mathbf{X}_{f,t-w}\right) \mathbf{A}'_c \mathbf{A}_c \left(\sum_{w \geq 0} a_{f,w} \mathbf{X}'_{f,t-w}\right) \mathbf{A}_{r,j},\right. \\ &\quad \left.\mathbf{A}'_{r,i} \cdot \left(\sum_{w \geq 0} a_{f,w} \mathbf{X}_{f,s-w}\right) \mathbf{A}'_c \mathbf{A}_c \left(\sum_{w \geq 0} a_{f,w} \mathbf{X}'_{f,s-w}\right) \mathbf{A}_{r,j}\right) \\ &= \frac{1}{\omega_r^2 T^2} \sum_{t=1}^T \sum_{w \geq 0} \sum_{l \geq 0} a_{f,w}^2 a_{f,l}^2 \cdot \text{Var}\left(\mathbf{A}'_{r,i} \cdot \mathbf{X}_{f,t-w} \mathbf{A}'_c \mathbf{A}_c \mathbf{X}'_{f,t-l} \mathbf{A}_{r,j}\right) \\ &= \frac{1}{\omega_r^2 T^2} \sum_{t=1}^T \sum_{w \geq 0} \sum_{l \geq 0} a_{f,w}^2 a_{f,l}^2 \cdot O(\|\mathbf{A}_c\|_F^4) = O(T^{-1} p^{-2\delta_{r,k_r}}) = o(1),\end{aligned}$$

where we used Assumption (F1) in the third last equality, and both (L1) and (F1) in the second last, which concludes (S.18). Then it holds that

$$\begin{aligned}&\left\| \frac{1}{\omega_r T} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}'_t - \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} \mathbf{A}_r \mathbf{A}'_r \right\|_F^2 \\ &\leq 2 \cdot \left\| \frac{1}{\omega_r T} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}'_t - \frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}'_t \right\|_F^2 + 2 \cdot \left\| \frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}'_t - \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} \mathbf{A}_r \mathbf{A}'_r \right\|_F^2 \\ &= 2 \cdot \left\| \frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{R}_{r,t} \right\|_F^2 + 2 \cdot \left\| \frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}'_t - \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} \mathbf{A}_r \mathbf{A}'_r \right\|_F^2 \\ &= O_P\left(T^{-1} p^{2(1-\delta_{r,k_r})} q^{1-2\delta_{c,1}} + p^{1-2\delta_{r,k_r}} q^{2(1-\delta_{c,1})}\right) + o_P(1) = o_P(1),\end{aligned}$$

where the second last equality used Lemma 2 and (S.18), and the last used Assumption (R1). Using the inequality that for the i -th eigenvalue of matrices $\widehat{\mathbf{A}}$ and $\mathbf{A}$, $|\lambda_i(\widehat{\mathbf{A}}) - \lambda_i(\mathbf{A})| \leq \|\widehat{\mathbf{A}} - \mathbf{A}\| \leq \|\widehat{\mathbf{A}} - \mathbf{A}\|_F$, we have for any $i \in [k_r]$,

$$|(\omega_r^{-1} \widehat{\mathbf{D}}_r)_{ii} - (\omega_r^{-1} \mathbf{D}_r)_{ii}| = o_P(1).$$

Thus, $\omega_r^{-1} \widehat{\mathbf{D}}_r \xrightarrow{P} \omega_r^{-1} \mathbf{D}_r$. This completes the proof of Lemma 4. $\square$

Lemma 5 (*Limit of $\mathbf{H}_r$ and $\mathbf{H}_c$*) Under Assumptions (F1), (L1), (E1), (E2), (R1) and (L2), we have

$$\begin{aligned}\mathbf{H}_r &\xrightarrow{P} \mathbf{H}_r^* := (\text{tr}(\mathbf{A}'_c \mathbf{A}_c))^{1/2} \cdot \mathbf{D}_r^{-1/2} (\mathbf{\Gamma}_r^*)' \mathbf{Z}_r^{1/2}, \\ \mathbf{H}_c &\xrightarrow{P} \mathbf{H}_c^* := (\text{tr}(\mathbf{A}'_r \mathbf{A}_r))^{1/2} \cdot \mathbf{D}_c^{-1/2} (\mathbf{\Gamma}_c^*)' \mathbf{Z}_c^{1/2},\end{aligned}$$

where $\mathbf{D}_r, \mathbf{D}_c$ are defined in Lemma 4, and $\mathbf{\Gamma}_r^*, \mathbf{\Gamma}_c^*$ are the eigenvector matrices of $\text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} \mathbf{Z}_r^{1/2} \mathbf{\Sigma}_{A,r} \mathbf{Z}_r^{1/2}$ and $\text{tr}(\mathbf{A}'_r \mathbf{A}_r) \cdot \omega_c^{-1} \mathbf{Z}_c^{1/2} \mathbf{\Sigma}_{A,c} \mathbf{Z}_c^{1/2}$, respectively.

Proof of Lemma 5. The proof of the two limits are similar, and hence we only show the probability limit of $\mathbf{H}_r$ is $\mathbf{H}_r^*$. First, left-multiply $\omega_r^{-1} \mathbf{Z}_r^{1/2} \mathbf{Q}'_r$ on (S.17), we have

$$\begin{aligned} (\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r) (\omega_r^{-1} \widehat{\mathbf{D}}_r) &= \omega_r^{-1} \mathbf{Z}_r^{1/2} \mathbf{Q}'_r \left(T^{-1} \sum_{t=1}^T \widehat{\mathbf{L}}_t \widehat{\mathbf{L}}'_t \right) \widehat{\mathbf{Q}}_r \\ &= \left[\frac{1}{T} \sum_{t=1}^T \omega_r^{-1} \mathbf{Z}_r^{1/2} \mathbf{Q}'_r \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}'_c \mathbf{Q}_c \mathbf{F}'_{Z,t} \mathbf{Z}_r^{-1/2} + \mathbf{R}_{r,\text{res}} \right] (\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r), \end{aligned}$$

where $\mathbf{R}_{r,\text{res}} := T^{-1} \sum_{t=1}^T \omega_r^{-1} \mathbf{Z}_r^{1/2} \mathbf{Q}'_r \mathbf{R}_{r,t} \widehat{\mathbf{Q}}_r (\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r)^{-1}$. This implies each column of $(\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r)$ is an eigenvector of the matrix $\left[T^{-1} \sum_{t=1}^T \omega_r^{-1} \mathbf{Z}_r^{1/2} \mathbf{Q}'_r \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}'_c \mathbf{Q}_c \mathbf{F}'_{Z,t} \mathbf{Z}_r^{-1/2} + \mathbf{R}_{r,\text{res}} \right]$. Note that

$$\begin{aligned} &\omega_r^{-1} \left[(\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r)' (\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r) - \text{tr}(\mathbf{A}'_c \mathbf{A}_c)^{-1} \cdot \mathbf{D}_r \right] \\ &= \omega_r^{-1} \widehat{\mathbf{Q}}' \mathbf{A}'_r \mathbf{A}_r \widehat{\mathbf{Q}} - \frac{1}{\text{tr}(\mathbf{A}'_c \mathbf{A}_c)} \widehat{\mathbf{Q}}' \left(\frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}'_t \right) \widehat{\mathbf{Q}} + \frac{1}{\text{tr}(\mathbf{A}'_c \mathbf{A}_c)} \widehat{\mathbf{Q}}' \left(\frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{C}_t \mathbf{C}'_t \right) \widehat{\mathbf{Q}} - \text{tr}(\mathbf{A}'_c \mathbf{A}_c)^{-1} \cdot \mathbf{D}_r, \end{aligned}$$

whose Frobenius norm is $o_P(1)$ by (S.18) and Lemma 4. We arrive at $(\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r)' (\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r) \xrightarrow{P} \text{tr}(\mathbf{A}'_c \mathbf{A}_c)^{-1} \cdot \mathbf{D}_r$. Thus, the eigenvalues of $(\mathbf{Q}'_r \widehat{\mathbf{Q}}_r)' (\mathbf{Q}'_r \widehat{\mathbf{Q}}_r)$ are asymptotically bounded away from zero and infinity by Assumption (L1), and hence $\|(\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r)^{-1}\|_F = O_P(\|\mathbf{Z}_r^{-1/2}\|_F)$. Therefore, we have

$$\begin{aligned} \|\mathbf{R}_{r,\text{res}}\|_F^2 &= O_P(1) \cdot \left\| \frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{R}_{r,t} \right\|_F^2 \cdot \|\mathbf{Z}_r^{1/2}\|_F^2 \cdot \|(\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r)^{-1}\|_F^2 \\ &= O_P \left(T^{-1} p^{2(1-\delta_r, k_r)} q^{1-2\delta_{c,1}} + p^{1-2\delta_{r, k_r}} q^{2(1-\delta_{c,1})} \right) = o_P(1), \end{aligned} \tag{S.19}$$

where we used Lemma 2 in the second last equality and Assumption (R1) in the last.

Denote by the following as the normalization of $\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r$,

$$\mathbf{\Gamma}_r := (\text{tr}(\mathbf{A}'_c \mathbf{A}_c))^{1/2} \cdot (\mathbf{Z}_r^{1/2} \mathbf{Q}'_r \widehat{\mathbf{Q}}_r) \mathbf{D}_r^{-1/2}.$$

Using the limit in (S.18), we have

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \omega_r^{-1} \mathbf{Z}_r^{1/2} \mathbf{Q}'_r \mathbf{Q}_r \mathbf{F}_{Z,t} \mathbf{Q}'_c \mathbf{Q}_c \mathbf{F}'_{Z,t} \mathbf{Z}_r^{-1/2} &= \frac{1}{\omega_r T} \sum_{t=1}^T \mathbf{A}'_r \mathbf{A}_r \mathbf{F}_t \mathbf{A}'_c \mathbf{A}_c \mathbf{F}'_t (\mathbf{A}'_r \mathbf{A}_r) (\mathbf{A}'_r \mathbf{A}_r)^{-1} \\ &\xrightarrow{P} \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} (\mathbf{A}'_r \mathbf{A}_r) (\mathbf{A}'_r \mathbf{A}_r) (\mathbf{A}'_r \mathbf{A}_r)^{-1} = \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} \mathbf{Z}_r^{1/2} \mathbf{\Sigma}_{A,r} \mathbf{Z}_r^{1/2}. \end{aligned}$$

By (S.19), Assumption (L2) and eigenvector perturbation theory, there exists a unique eigenvector matrix $\mathbf{\Gamma}_r^*$ of $\text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \omega_r^{-1} \mathbf{Z}_r^{1/2} \mathbf{\Sigma}_{A,r} \mathbf{Z}_r^{1/2}$ such that $\|\mathbf{\Gamma}_r^* - \mathbf{\Gamma}_r\| = o_P(1)$. Thus,

$$\widehat{\mathbf{Q}}'_r \mathbf{Q}_r = (\text{tr}(\mathbf{A}'_c \mathbf{A}_c))^{-1/2} \mathbf{D}_r^{1/2} \mathbf{\Gamma}_r' \mathbf{Z}_r^{-1/2} \xrightarrow{P} (\text{tr}(\mathbf{A}'_c \mathbf{A}_c))^{-1/2} \mathbf{D}_r^{1/2} (\mathbf{\Gamma}_r^*)' \mathbf{Z}_r^{-1/2},$$

and hence using (S.18) again, we obtain

$$\begin{aligned} \mathbf{H}_r &= \widehat{\mathbf{D}}_r^{-1} \widehat{\mathbf{Q}}'_r \left(\frac{1}{T} \sum_{t=1}^T \mathbf{A}_r \mathbf{F}_t \mathbf{A}'_c \mathbf{A}_c \mathbf{F}'_t \mathbf{A}'_r \right) \mathbf{Q}_r \mathbf{\Sigma}_{A,r}^{-1} \xrightarrow{P} \mathbf{D}_r^{-1} \widehat{\mathbf{Q}}'_r \left(\text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \mathbf{A}_r \mathbf{A}'_r \right) \mathbf{Q}_r \mathbf{\Sigma}_{A,r}^{-1} \\ &= \text{tr}(\mathbf{A}'_c \mathbf{A}_c) \cdot \mathbf{D}_r^{-1} \widehat{\mathbf{Q}}'_r \mathbf{Q}_r \mathbf{Z}_r^{1/2} \mathbf{Z}_r^{1/2} \mathbf{Q}'_r \mathbf{Q}_r \mathbf{\Sigma}_{A,r}^{-1} \xrightarrow{P} (\text{tr}(\mathbf{A}'_c \mathbf{A}_c))^{1/2} \cdot \mathbf{D}_r^{-1/2} (\mathbf{\Gamma}_r^*)' \mathbf{Z}_r^{1/2}, \end{aligned}$$

which completes the proof of Lemma 5. $\square$

Lemma 6 Under the assumptions in Theorem 3, for any $j \in [p], l \in [q]$,

$$\|\hat{\mathbf{Q}}_{r,j} - \mathbf{H}_r \mathbf{Q}_{r,j}\|^2 = O_P(T^{-1} p^{\delta_{r,1}-2\delta_{r,k_r}} q^{1-2\delta_{c,1}} + p^{-3\delta_{r,k_r}} q^{2-2\delta_{c,1}}), \quad (\text{S.20})$$

$$\|\hat{\mathbf{Q}}_{c,l} - \mathbf{H}_c \mathbf{Q}_{c,l}\|^2 = O_P(T^{-1} q^{\delta_{c,1}-2\delta_{c,k_c}} p^{1-2\delta_{r,1}} + q^{-3\delta_{c,k_c}} p^{2-2\delta_{r,1}}). \quad (\text{S.21})$$

Proof of Lemma 6. For $\hat{\mathbf{Q}}_{r,j} - \mathbf{H}_r \mathbf{Q}_{r,j}$, first consider the case when $Tq = o(p^{\delta_{r,k_r}+\delta_{r,1}})$. From (S.27) in the proof of Theorem 5, we have

$$\|\hat{\mathbf{Q}}_{r,j} - \mathbf{H}_r \mathbf{Q}_{r,j}\|^2 = O_P((Tp^{2\delta_{r,k_r}-\delta_{r,1}} q^{2\delta_{c,1}-1})^{-1}). \quad (\text{S.22})$$

Now suppose $Tq = o(p^{\delta_{r,k_r}+\delta_{r,1}})$ fails to hold. From the decomposition of $\hat{\mathbf{Q}}_{r,j} - \mathbf{H}_r \mathbf{Q}_{r,j}$ in (S.21), the leading term among the expressions will be $\mathcal{I}_3$. It has rate

$$\|\mathcal{I}_3\|^2 = O_P(T^{-1} p^{-2\delta_{r,k_r}} q^{1-2\delta_{c,1}} + p^{-3\delta_{r,k_r}} q^{2-2\delta_{c,1}}) = O_P(p^{-3\delta_{r,k_r}} q^{2-2\delta_{c,1}}),$$

where the last equality used the fact that $Tq = o(p^{\delta_{r,k_r}+\delta_{r,1}})$ does not hold. Thus we have

$$\|\hat{\mathbf{Q}}_{r,j} - \mathbf{H}_r \mathbf{Q}_{r,j}\|^2 = O_P(p^{-3\delta_{r,k_r}} q^{2-2\delta_{c,1}}). \quad (\text{S.23})$$

Combining (S.22) and (S.23), we arrive at the statement of (S.20). The proof for (S.21) is similar and omitted here, which ends of the proof of Lemma 6. $\square$

Lemma 7 Let Assumptions (F1), (E1), (E2) and (E3) hold. Then for any deterministic vectors $\mathbf{u} \in \mathbb{R}^{k_r}$ and $\mathbf{v} \in \mathbb{R}^{k_c}$ with constant magnitudes,

$$\max_{j \in [p]} \frac{1}{(qT)^{1/2}} \sum_{h=1}^q \sum_{t=1}^T E_{t,jh} \mathbf{u}' \mathbf{F}_t \mathbf{v} = O_P(\log(p)), \quad \max_{i \in [q]} \frac{1}{(pT)^{1/2}} \sum_{h=1}^p \sum_{t=1}^T E_{t,hi} \mathbf{v}' \mathbf{F}_t' \mathbf{u} = O_P(\log(q)).$$

Proof of Lemma 7. We only show the first result as the proof for the second is similar. From Assumption (F1), (E1) and (E2), we may rewrite $E_{t,jh}$ and $\mathbf{u}' \mathbf{F}_t \mathbf{v}$ as

$$\begin{aligned} E_{t,jh} &= \mathbf{A}'_{e,r,j} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,h} + \Sigma_{\epsilon,jh} \left(\sum_{g \geq 0} a_{\epsilon,g} X_{\epsilon,t-g,jh} \right), \\ \mathbf{u}' \mathbf{F}_t \mathbf{v} &= \sum_{m=1}^{k_r} \sum_{n=1}^{k_c} \sum_{l \geq 0} a_{f,l} X_{f,t-l,mn} u_m v_n, \end{aligned}$$

so that for any $j \in [p]$,

$$\begin{aligned} \sum_{h=1}^q \sum_{t=1}^T E_{t,jh} \mathbf{u}' \mathbf{F}_t \mathbf{v} &= \sum_{h=1}^q \sum_{t=1}^T \mathbf{A}'_{e,r,j} \left(\sum_{w \geq 0} a_{e,w} \mathbf{X}_{e,t-w} \right) \mathbf{A}_{e,c,h} \sum_{m=1}^{k_r} \sum_{n=1}^{k_c} \sum_{l \geq 0} a_{f,l} X_{f,t-l,mn} u_m v_n \\ &\quad + \sum_{h=1}^q \sum_{t=1}^T \Sigma_{\epsilon,jh} \left(\sum_{g \geq 0} a_{\epsilon,g} X_{\epsilon,t-g,jh} \right) \sum_{m=1}^{k_r} \sum_{n=1}^{k_c} \sum_{l \geq 0} a_{f,l} X_{f,t-l,mn} u_m v_n. \end{aligned} \quad (\text{S.24})$$

Consider first the second term above, i.e.,

$$\sum_{m=1}^{k_r} \sum_{n=1}^{k_c} u_m v_n \sum_{l \geq 0} \sum_{g \geq 0} a_{f,l} a_{\epsilon,g} \sum_{t=1}^T \left(\sum_{h=1}^q \Sigma_{\epsilon,jh} X_{\epsilon,t-g,jh} \right) X_{f,t-l,mn}.$$

Fix $l \geq 0, g \geq 0$. By Assumption (E3), for each $t \in [T]$, we have $\sum_{h=1}^q \Sigma_{\epsilon,jh} X_{\epsilon,t-g,jh} \sim \text{subG}(C_1 q)$, with arbitrary constant $C_1 > 0$ such that $C_1 q = \sum_{h=1}^q \Sigma_{\epsilon,jh}^2$, which is independent over g . Notice that

$X_{f,t-l,mn} \sim \text{subG}(1)$ by (E3), then $(\sum_{h=1}^q \Sigma_{\epsilon,jh} X_{\epsilon,t-g,jh}) X_{f,t-l,mn} \sim \text{subE}(\sqrt{C_1 q})$ which is independent over t , and hence $\sum_{t=1}^T (\sum_{h=1}^q \Sigma_{\epsilon,jh} X_{\epsilon,t-g,jh}) X_{f,t-l,mn} \sim \text{subE}(\sqrt{C_1 q T})$. Then sum those sub-exponential random variables over $l \geq 0, g \geq 0$, we have by (E2),

$$\sum_{l \geq 0} \sum_{g \geq 0} a_{f,l} a_{\epsilon,g} \sum_{t=1}^T \left(\sum_{h=1}^q \Sigma_{\epsilon,jh} X_{\epsilon,t-g,jh} \right) X_{f,t-l,mn} \sim \text{subE}(\sqrt{C_2 q T}),$$

with some arbitrary constant $C_2 > 0$. As k_r, k_c, k_{e_r} and $k_{e,c}$ are all constants, we conclude that the entire second term in (S.24), together with the first term therein, are also sub-exponential with parameter of the rate $\sqrt{qT}$. Therefore, for each $j \in [p]$, it holds that $(qT)^{-1/2} \sum_{h=1}^q \sum_{t=1}^T E_{t,jh} \mathbf{u}' \mathbf{F}_t \mathbf{v}$ is sub-exponential with parameter of constant rate. Using the union bound, with some arbitrary constant $C_3 > 0$, we have

$$\mathbb{P} \left(\max_{j \in [p]} \frac{1}{(qT)^{1/2}} \sum_{h=1}^q \sum_{t=1}^T E_{t,jh} \mathbf{u}' \mathbf{F}_t \mathbf{v} \geq \varepsilon \right) \leq \exp \{ \log(p) - C_3 \varepsilon \}, \quad (\text{S.25})$$

implying that $\max_{j \in [p]} (qT)^{-1/2} \sum_{h=1}^q \sum_{t=1}^T E_{t,jh} \mathbf{u}' \mathbf{F}_t \mathbf{v} = O_P(\log(p))$. This concludes the proof for the lemma. $\square$

Lemma 8 *Let all assumptions in Theorem 3 hold, and let Assumption (E3) also hold. Then with $\mathbf{R}_{r,t}$ defined in (S.15) and $\mathbf{R}_{c,t}$ in (S.16), we have*

$$\max_{j \in [p]} \left\| \left(\sum_{t=1}^T \mathbf{R}_{r,t} \right)_{\cdot j} \right\|^2 = O_P((Tpq + T^2 q^2) \log^2(p)), \quad (\text{S.26})$$

$$\max_{j \in [q]} \left\| \left(\sum_{t=1}^T \mathbf{R}_{c,t} \right)_{\cdot j} \right\|^2 = O_P((Tpq + T^2 p^2) \log^2(q)), \quad (\text{S.27})$$

Proof of Lemma 8. Consider (S.26). Essentially, we need to show similar results in Lemma 2. To this end, we show the corresponding versions of (S.1) and (S.2), and the remaining terms are based on the derived results and can be shown (omitted here) using the same machinery of sub-exponential distribution and the union bound as in (S.25). To start with, using the first result in Lemma 7, we have

$$\max_{j \in [p]} \left\| \left(\sum_{t=1}^T \mathbf{C}_t \mathbf{E}'_t \right)_{\cdot j} \right\|_F^2 = \sum_{i=1}^p \|\mathbf{A}_{r,i}\|^2 \cdot \max_{l \in [p]} \left(\sum_{h=1}^q \sum_{t=1}^T E_{t,lh} \frac{1}{\|\mathbf{A}_{r,i}\|} \mathbf{A}'_{r,i} \mathbf{F}_t \mathbf{A}_{c,h} \right)^2 = O_P(Tp^{\delta_{r,1}} q \log^2(p)).$$

Similarly,

$$\begin{aligned} & \max_{j \in [p]} \left\| \left(\sum_{t=1}^T \mathbf{E}_t \mathbf{C}'_t \right)_{\cdot j} \right\|_F^2 = \max_{i \in [p]} \left\{ \|\mathbf{A}_{r,i}\|^2 \cdot \sum_{l=1}^p \left(\sum_{h=1}^q \sum_{t=1}^T E_{t,lh} \frac{1}{\|\mathbf{A}_{r,i}\|} \mathbf{A}'_{r,i} \mathbf{F}_t \mathbf{A}_{c,h} \right)^2 \right\} \\ & \leq \max_{i \in [p]} \|\mathbf{A}_{r,i}\|^2 \cdot \max_{i \in [p]} \sum_{l=1}^p \left(\sum_{h=1}^q \sum_{t=1}^T E_{t,lh} \frac{1}{\|\mathbf{A}_{r,i}\|} \mathbf{A}'_{r,i} \mathbf{F}_t \mathbf{A}_{c,h} \right)^2 = O_P(Tpq \log^2(p)). \end{aligned}$$

Next, consider

$$\max_{j \in [p]} \left\| \sum_{t=1}^T (\mathbf{E}_t \mathbf{E}'_t)_{\cdot j} \right\|_F^2 = \max_{j \in [p]} \sum_{i=1}^p \left(\sum_{t=1}^T \sum_{h=1}^q E_{t,ih} E_{t,jh} \right)^2 \quad (\text{S.28})$$

Given $j \in [p]$, first consider first $i \neq j$. By Assumption (E1) and (E2), we can write

$$\begin{aligned} E_{t,jh} &= \sum_{m=1}^{k_{e,r}} \sum_{n=1}^{k_{e,c}} A_{e,r,jm} A_{e,c,hn} \left(\sum_{w \geq 0} a_{e,w} X_{e,t-w,mn} \right) + \Sigma_{\epsilon,jh} \left(\sum_{g \geq 0} a_{\epsilon,g} X_{\epsilon,t-g,jh} \right), \\ E_{t,ih} &= \sum_{\tau=1}^{k_{e,r}} \sum_{\gamma=1}^{k_{e,c}} A_{e,r,i\tau} A_{e,c,h\gamma} \left(\sum_{l \geq 0} a_{e,l} X_{e,t-l,\tau\gamma} \right) + \Sigma_{\epsilon,ih} \left(\sum_{\xi \geq 0} a_{\epsilon,\xi} X_{\epsilon,t-\xi,ih} \right). \end{aligned}$$

Then among all terms in the expansion of $\sum_{t=1}^T \sum_{h=1}^q E_{t,ih} E_{t,jh}$, consider

$$\sum_{g \geq 0} \sum_{\xi \geq 0} a_{\epsilon,g} a_{\epsilon,\xi} \sum_{t=1}^T \sum_{h=1}^q \Sigma_{\epsilon,jh} \Sigma_{\epsilon,ih} X_{\epsilon,t-g,jh} X_{\epsilon,t-\xi,ih}.$$

Fix $g \geq 0$ and $\xi \geq 0$, then it is direct from Assumption (E3) that $\Sigma_{\epsilon,jh} \Sigma_{\epsilon,ih} X_{\epsilon,t-g,jh} X_{\epsilon,t-\xi,ih}$ is sub-exponential with parameter of constant order and independent over $h \in [q]$ and $t \in [T]$. This implies $\sum_{t=1}^T \sum_{h=1}^q \Sigma_{\epsilon,jh} \Sigma_{\epsilon,ih} X_{\epsilon,t-g,jh} X_{\epsilon,t-\xi,ih}$ is sub-exponential with parameter of order $(Tq)^{1/2}$, which hence also holds true for $\sum_{g \geq 0} \sum_{\xi \geq 0} a_{\epsilon,g} a_{\epsilon,\xi} \sum_{t=1}^T \sum_{h=1}^q \Sigma_{\epsilon,jh} \Sigma_{\epsilon,ih} X_{\epsilon,t-g,jh} X_{\epsilon,t-\xi,ih}$ by Assumption (E2). Thus,

$$\max_{j \in [p]} \sum_{i \neq j}^p \left\{ \sum_{t=1}^T \sum_{h=1}^q \Sigma_{\epsilon,jh} \left(\sum_{g \geq 0} a_{\epsilon,g} X_{\epsilon,t-g,jh} \right) \Sigma_{\epsilon,ih} \left(\sum_{\xi \geq 0} a_{\epsilon,\xi} X_{\epsilon,t-\xi,ih} \right) \right\}^2 = O_P(Tpq \log^2(p)). \quad (\text{S.29})$$

Using the same argument above, with the independence between $\{\mathbf{X}_{e,t}\}$ and $\{\mathbf{X}_{\epsilon,t}\}$ from (E2), we have

$$\begin{aligned} & \max_{j \in [p]} \sum_{i=1}^p \left\{ \sum_{t=1}^T \sum_{h=1}^q \sum_{m=1}^{k_{e,r}} \sum_{n=1}^{k_{e,c}} A_{e,r,jm} A_{e,c,hn} \left(\sum_{w \geq 0} a_{e,w} X_{e,t-w,mn} \right) \Sigma_{\epsilon,ih} \left(\sum_{\xi \geq 0} a_{\epsilon,\xi} X_{\epsilon,t-\xi,ih} \right) \right\}^2 \\ &= O_P(Tpq \log^2(p)), \\ & \max_{j \in [p]} \sum_{i=1}^p \left\{ \sum_{t=1}^T \sum_{h=1}^q \Sigma_{\epsilon,jh} \left(\sum_{g \geq 0} a_{\epsilon,g} X_{\epsilon,t-g,jh} \right) \sum_{\tau=1}^{k_{e,r}} \sum_{\gamma=1}^{k_{e,c}} A_{e,r,i\tau} A_{e,c,h\gamma} \left(\sum_{l \geq 0} a_{e,l} X_{e,t-l,\tau\gamma} \right) \right\}^2 \\ &= O_P(Tpq \log^2(p)). \end{aligned} \quad (\text{S.30})$$

In the expansion of $\sum_{t=1}^T \sum_{h=1}^q E_{t,ih} E_{t,jh}$ for any $i \in [p]$, consider now

$$\sum_{w \geq 0} \sum_{l \geq 0} a_{e,w} a_{e,l} \sum_{t=1}^T \sum_{h=1}^q \sum_{m=1}^{k_{e,r}} \sum_{n=1}^{k_{e,c}} \sum_{\tau=1}^{k_{e,r}} \sum_{\gamma=1}^{k_{e,c}} A_{e,r,i\tau} A_{e,c,h\gamma} A_{e,r,jm} A_{e,c,hn} X_{e,t-w,mn} X_{e,t-l,\tau\gamma},$$

which is sub-exponential with mean of order T and parameter of order $\|\mathbf{A}_{e,r}\|_\infty \cdot (T)^{1/2}$ by Assumption (E1), (E2) and (E3). Hence by the sparsity of $\mathbf{A}_{e,r}$ according to (E1) again,

$$\begin{aligned} & \max_{j \in [p]} \sum_{i=1}^p \left\{ \sum_{t=1}^T \sum_{h=1}^q \sum_{m=1}^{k_{e,r}} \sum_{n=1}^{k_{e,c}} A_{e,r,jm} A_{e,c,hn} \left(\sum_{w \geq 0} a_{e,w} X_{e,t-w,mn} \right) \right. \\ & \left. \sum_{\tau=1}^{k_{e,r}} \sum_{\gamma=1}^{k_{e,c}} A_{e,r,i\tau} A_{e,c,h\gamma} \left(\sum_{l \geq 0} a_{e,l} X_{e,t-l,\tau\gamma} \right) \right\}^2 = O_P(T^2 \log^2(p)). \end{aligned} \quad (\text{S.31})$$

To bound (S.28), it remains to consider

$$\sum_{g \geq 0} \sum_{\xi \geq 0} a_{\epsilon,g} a_{\epsilon,\xi} \sum_{t=1}^T \sum_{h=1}^q \Sigma_{\epsilon,jh}^2 X_{\epsilon,t-g,jh}^2,$$

which is sub-exponential with parameter of order $(Tq)^{1/2}$, similar to the case as in (S.29), except that the mean is of order Tq . Therefore,

$$\max_{j \in [p]} \mathbb{1}\{i = j\} \left\{ \sum_{t=1}^T \sum_{h=1}^q \Sigma_{\epsilon,jh} \left(\sum_{g \geq 0} a_{\epsilon,g} X_{\epsilon,t-g,jh} \right) \Sigma_{\epsilon,ih} \left(\sum_{\xi \geq 0} a_{\epsilon,\xi} X_{\epsilon,t-\xi,ih} \right) \right\}^2 = O_P(T^2 q^2 \log^2(p)). \quad (\text{S.32})$$

Finally for (S.28), combining (S.29), (S.30), (S.31) and (S.32), we have

$$\max_{j \in [p]} \left\| \sum_{t=1}^T (\mathbf{E}_t \mathbf{E}_t')_{\cdot j} \right\|_F^2 = O_P((Tpq + T^2 q^2) \log^2(p)).$$

This ends the proof for (S.26). The result (S.27) follows similarly to (S.26) and this concludes the proof of the lemma. $\square$

Lemma 9 *Let all assumptions in Theorem 3 hold, and let Assumption (E3) also hold. Then we have*

$$\max_{j \in [p]} \|\widehat{\mathbf{Q}}_{r,j\cdot} - \mathbf{H}_r \mathbf{Q}_{r,j\cdot}\|^2 = O_P \left\{ (T^{-1} p^{1-2\delta_{r,k_r}}) q^{1-2\delta_{c,1}} + p^{-2\delta_{r,k_r}} q^{2(1-\delta_{c,1})} \log^2(p) \right\}, \quad (\text{S.33})$$

$$\max_{j \in [q]} \|\widehat{\mathbf{Q}}_{c,j\cdot} - \mathbf{H}_c \mathbf{Q}_{c,j\cdot}\|^2 = O_P \left\{ (T^{-1} q^{1-2\delta_{c,k_c}}) p^{1-2\delta_{r,1}} + q^{-2\delta_{c,k_c}} p^{2(1-\delta_{r,1})} \log^2(q) \right\}, \quad (\text{S.34})$$

$$\begin{aligned} & \max_{t \in [T]} \|\widehat{\mathbf{F}}_{Z,t} - (\mathbf{H}_r^{-1})' \mathbf{F}_{Z,t} \mathbf{H}_c^{-1}\|_F^2 \\ &= O_P \left\{ (p^{1-\delta_{r,k_r}} q^{1-\delta_{c,k_c}} + T^{-1} p^{1+2\delta_{r,1}-2\delta_{r,k_r}} q^{1-\delta_{c,1}} + p^{1+\delta_{r,1}-3\delta_{r,k_r}} q^{2-\delta_{c,1}} \right. \\ & \quad \left. + T^{-1} q^{1+2\delta_{c,1}-2\delta_{c,k_c}} p^{1-\delta_{r,1}} + q^{1+\delta_{c,1}-3\delta_{c,k_c}} p^{2-\delta_{r,1}}) \log^2(T) \right\}. \end{aligned} \quad (\text{S.35})$$

Thus, we have

$$\begin{aligned} & \max_{i \in [p], j \in [q], t \in [T]} (\widehat{C}_{t,ij} - C_{t,ij})^2 \\ &= O_P \left\{ (p^{1-2\delta_{r,k_r}} q^{1-2\delta_{c,k_c}} + T^{-1} p^{1+2\delta_{r,1}-3\delta_{r,k_r}} q^{1-\delta_{c,1}-\delta_{c,k_c}} + p^{1+\delta_{r,1}-4\delta_{r,k_r}} q^{2-\delta_{c,1}-\delta_{c,k_c}} \right. \\ & \quad \left. + T^{-1} q^{1+2\delta_{c,1}-3\delta_{c,k_c}} p^{1-\delta_{r,1}-\delta_{r,k_r}} + q^{1+\delta_{c,1}-4\delta_{c,k_c}} p^{2-\delta_{r,1}-\delta_{r,k_r}}) \log^2(T) \log^2(p) \log^2(q) \right\}. \end{aligned}$$

Proof of Lemma 9. To see (S.33), from the proof of (4.8) in Theorem 2, we have

$$\begin{aligned} & \max_{j \in [p]} \|\widehat{\mathbf{Q}}_{r,j\cdot} - \mathbf{H}_r \mathbf{Q}_{r,j\cdot}\|^2 = \max_{j \in [p]} \left\| T^{-1} \widehat{\mathbf{D}}_r^{-1} \widehat{\mathbf{Q}}_r' \left(\sum_{t=1}^T \mathbf{R}_{r,t} \right)_{\cdot j} \right\|^2 \\ & \leq T^{-2} \cdot \|\widehat{\mathbf{D}}_r^{-1}\|_F^2 \cdot \|\widehat{\mathbf{Q}}_r\|_F^2 \cdot \max_{j \in [p]} \left\| \left(\sum_{t=1}^T \mathbf{R}_{r,t} \right)_{\cdot j} \right\|_F^2 \\ & = O_P \left(T^{-1} p^{1-2\delta_{r,k_r}} q^{1-2\delta_{c,1}} \log^2(p) + p^{-2\delta_{r,k_r}} q^{2(1-\delta_{c,1})} \log^2(p) \right), \end{aligned}$$

where the last equality used Lemma 3 and Lemma 8. In a similar way, (S.34) holds also by Lemma 3 and Lemma 8.

For (S.35), by inspecting (S.18), it suffices to characterize the change from the rate of $\|\mathbf{F}_t\|_F^2$ to the rate of $\max_{t \in [T]} \|\mathbf{F}_t\|_F^2$, while all other rates follow the similar arguments in the proof of Lemma 8 by using sub-exponential distributions. With Assumption (E3), $\|\mathbf{F}_t\|_F^2$ is sub-exponential with both mean and parameter of constant order, so that $\max_{t \in [T]} \|\mathbf{F}_t\|_F^2 = O_P(\log^2(T))$. By checking all the rates in the expansion (S.18) are inflated by $\log^2(T)$, (S.35) is hence concluded.

Finally, with all previous results and recall the expansion in (S.20), the rate of $\max_{i \in [p], j \in [q], t \in [T]} (\widehat{C}_{t,ij} - C_{t,ij})^2$ is inflated by $\log^2(T) \log^2(p) \log^2(q)$ compared to the individual rate of $(\widehat{C}_{t,ij} - C_{t,ij})^2$. This ends the proof of the lemma. $\square$

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